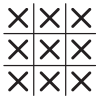
PERFECT PROCESSES



RUNE JOSEFSEN
KRISTIAN STEEN HOLME

PERFECT PROCESSES

PRAKTISK FORLAG

Perfect Processes
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Cover: Marta Podolska
Illustrations: Rasmus Steen Rasmussen
Translation: Andrew Culligan
Editor: Lucy Hogan
Revised edition 2016
ISBN 978-87-996855-2-3

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FOREWORD

This book has been in development for a long time. We based it on our experiences working for a wide range of organisations in Denmark and abroad over the last 20 years.

The title of the book, *Perfect Processes*, does not refer to perfection in the strict sense. In fact, the closer we feel we are to perfection in a process, the farther away from it we actually are. Reaching this goal is not a static end point, but an on-going effort by an organisation to constantly search for better and more efficient ways of doing things.

Thanks are due to too many people to mention for helping, inspiring and challenging us along the way, thereby improving the end result in many respects.

We hope that this book can help you in your work with process improvement.

Rune & Kristian

ABOUT THE AUTHORS

Rune Josefsen
Rune has more than 20 years' experience as a process consultant for numerous companies in Denmark, Europe, Asia, the Middle East and Africa. He previously worked as a senior consultant and manager at Accenture and IBM, and today works as a senior partner in Kvalitor Consulting, which specialises in training and project implementation in the field of process improvement. Rune holds an MScSoc in economics and communications from the University of Roskilde, Denmark, and has studied at the University of Bath, the United Kingdom, and L'Université Stendhal, France.

Kristian Steen Holme
Kristian has worked in leading roles with data-driven optimisation processes at A.P. Moller-Maersk and has implemented Lean in the administration of, among others, the Danish National Police and the R&D Department at Siemens Wind Power. Today Kristian works as an independent management and strategy consultant. Kristian holds an MSc in Business and IT from Copenhagen Business School and has also studied organisational change and international business at MBA level at Simon Fraser University, Canada.

1 · INTRODUCTION

What is a process? There are many definitions of a process, but fundamentally it is a series of activities we execute in order to achieve a desired result. Processes will almost always involve various employees and departments, and this is often part of the reason why they do not function optimally.

The purpose of this book is to help you to start creating better processes in your organisation, whether you are a senior executive, a mid-level manager, a project leader or an employee. We will go through how you can work on process improvement as a project (chaps 2 to 5) or as a part of daily operations (chap. 6).

Our starting position when working on process optimisation is the recognition that even though we do a lot right, we do not always work in the best possible way. We often spend too much time, use too many resources, make errors or focus on unimportant aspects in our processes. This means we can be working very hard and feel that we are making a huge effort without necessarily satisfying our customers. So how are perfect processes created, as the title of our book promises? The answer, as noted in our foreword, is that a perfect process does not exist. Processes must be maintained and continually adjusted and improved upon. The process of perfecting never ends.

Process optimisation is about looking at whether we solve tasks in the right way and if we solve the right tasks. Meaning, it is not about hassling employees to work faster, but rather about working smarter so your customers and your organisation get the most from the effort invested.

Even though a lot of process optimisation is based on common sense, it is often advantageous to systematise these common-sense

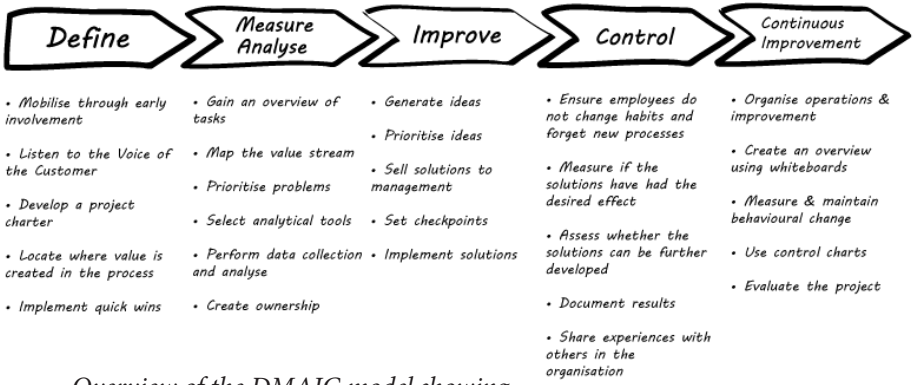
insights, so you don't have to start from scratch every time but can instead learn from others who have wrestled with the same challenges. In this book, therefore, we will work with two of the most widely used methods of process optimisation at present: Lean and Six Sigma.

Lean and Six Sigma are methods of analysing and improving processes. Each has its pros and cons, however, from our experience in process improvement projects, we know that they work together effectively and complement each other.

We have chosen to model this book on the five phases of a Six Sigma project:

1. We begin by defining the task.
2. Then we gather data.
3. Next we analyse it.
4. We implement solutions.
5. Finally, we follow up on the success or otherwise of the implementation.

There are many different models for the phases of this work but we think that the Six Sigma model can be intuitively understood by most people and is easy to work with.



Overview of the DMAIC model showing project phases and the most important project activities.

What is Lean?

Lean builds on the principles developed primarily by the Toyota Production System (TPS) since the 1960s. TPS was Toyota's internal productivity improvement programme and was partially inspired by Training within Industry (TWI), which was developed by the United States Department of War during the Second World War to improve the productivity of suppliers of essential war materials.

The philosophy of Lean stems largely from the issue of limited access to resources. After the Second World War, Toyota, unlike some of its competitors, was not in a position to invest in large warehouses and production facilities on different continents. This led to the implementation of 'Just-in-Time', a system whereby the raw materials arrive exactly when they are needed, and where parts are produced just before they are to be used, thereby minimising the inventory held throughout the production process. Moreover, Toyota took into account that even the smallest error could have fatal consequences for its success in the market. This inbuilt caution is credited with being one of the most important factors in Toyota's success.

Lean's ideal is the 'one piece flow'. This means that an employee carries out and completes a task (whether in production, reporting, processing purchase orders or responding to customer queries) before beginning a new task. Lean is therefore sometimes described as the opposite to mass production where producing as many products as possible of primary importance. These products will then be shelved for later use. However, if a customer changes their order or orders to something non-standard, the company must deliver an item that is not in stock and those shelved goods remain in the warehouse, growing obsolete.

In relation to process design, the important thing is to be able to redirect processes rapidly so that a company can deliver what a customer needs. If the company has large stocks of outdated products, it may be difficult or even impossible to sell them, especially when its customers have gone elsewhere.

The fundamental idea of Lean is to analyse the time spent delivering a product or service and to cut back on everything that does not create or add value for the end customer — called waste (*Muda* in Japanese). The result is a quicker process with reduced resource usage.

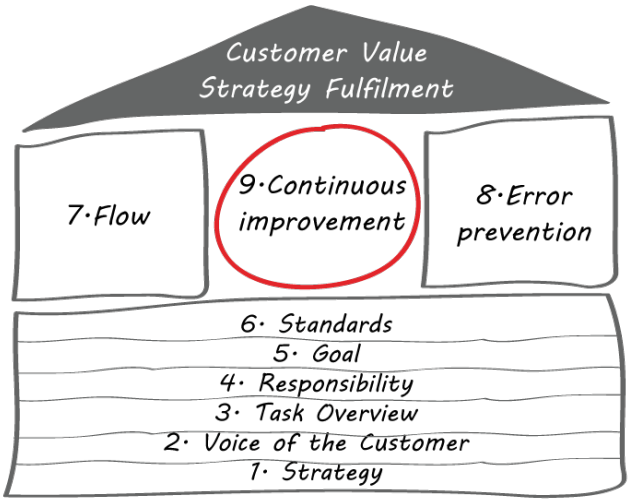
Another important principle of Lean is the close involvement of employees in problem-solving. Employees are entrusted with the responsibility of not just carrying out their appointed tasks but continually trying to improve on their work processes. Some of the ideas which came out of this revolved around quality circles, a concept that was introduced by Kaoru Ishikawa in the 1960s. They consist of a number of workers continuously working on solving problems in relation to suboptimal processes so as to improve operations.

Working with the above Lean principles is often summarised as follows:

1. **Identify what is of value to the customer**. Start by creating an understanding of what the customer values. Understanding this is essential for optimising processes and requires active listening.
2. **Map the value stream and eliminate waste**. Gain an overview of where time is spent, where projects come to a standstill, and where time is spent on activities that do not add value for the customer.
3. **Create flow**. Flow is the way that the case or product moves quickly through the process and does not stop along the way. This creates a reduced inventory, faster and more flexible production, and fewer errors.

4. **Implement a pull system**. In a pull system, an activity first takes place upon demand from the next link in the chain; production does not create an inventory, but only responds to demand.
5. **Engage in continuous improvement**. The last step is about continually improving processes and this is the responsibility of every person in an organisation.

Lean also works with a series of principles which must take place in a specific sequence to achieve success. A house or temple is often used to illustrate the principles which are foundational for the implementation of more advanced ones. The authors' experiences in implementing Lean can be summed up by this Lean house which details the five principles, in particular the basic phases which make up the foundation.



The Lean house gives an overview of Lean's focus areas and the sequence of improvement initiatives.

Strategy:

It is essential that the organisation's strategy is clear before you begin to optimise processes. Do you want to supply the cheapest products? Do you want to be known for your rapid service? What position do you want to occupy in the market? This will set the overall objectives of your process optimisation and determine what you should prioritise.

The customer's voice:

'The customer's voice' is your customers' view of the company's products and services, and their demands in this regard. You need to understand exactly what your customers value and what they don't value. Are you disappointing your customers today by not delivering something they need? Are you wasting resources in supplying services or products that the customers don't value? An understanding of what has value for customers is the basis of all the following activities. Just think of the effort you waste providing customers with something they don't want or need, or how poor their experience is when their wishes are not understood.

Task overview:

Part of the foundation of process optimisation is creating an overview of the current situation. Which tasks do we spend time on and in what proportion? Could some tasks be combined advantageously? This overview will be extremely important in subsequent optimisation. For example, it will be used when prioritising the areas to be worked on and deciding upon where to begin. We will also use it to document our baseline so we can see if we have become quicker and better or whether the changes implemented have decreased efficiency. It will also be used as input to improve the daily direction of the work.

Responsibility:

Processes always involve numerous functions or departments within an organisation and problems can arise due to ineffective or non-existent interdepartmental cooperation, or a lack of clarity as to how responsibility is shared. The aim is to ensure that clear rules for co-operation and handovers in the process are in place: who is responsible for the deliverables in the process? What is the acceptable delivery time? How is delivery to take place? What information must accompany the delivery?

Goals:

The next element in the foundation must ensure that there are well-defined goals for everyone in the organisation and continual feedback on performance related to those goals. If expectations regarding employees' performance and goals are unclear, it can be stressful for the employee and difficult for them to know when the goal has been reached.

Standards:

The foundation is completed by establishing relevant standards. This may take the form of work procedures, checklists, colour coding, standard letters, designated places for tools, etc. It could also involve ensuring that desks, workstations and workshops are orderly and clean so that time is not wasted unnecessarily in searching for tools, materials and so on. Standards ensure that you can achieve quality and efficiency without being experts in everything.

Flow:

Once you have laid the foundation, it becomes possible to begin on the next level of the Lean house. As mentioned above, flow is about shortening the time it takes a product or case to move through the

process. Simply put, aim for the shortest possible time from the customer expressing a need to that need being fulfilled.

Prevent errors:

Mistakes waste resources and create frustration right across an organisation. They are also often the cause of extra time spent on checks and controls, and can delay work and destroy flow. Our view is that it is okay to make mistakes, but not to continue making the same mistake. Working on error prevention helps you to discover the underlying cause and to take steps to ensure that the mistake does not arise again.

Continuous improvement:

Continuous improvement means making optimisation an integral part of daily operations, an element of the organisation's culture and not simply a project with a deadline. Try to motivate employees to continually identify and suggest changes so that they can constantly improve.

Follow your strategy and deliver customer value:

The roof of the Lean house ensures that all the activities set in motion aim to create value for the customer and fulfil the organisation's strategy. Working with process optimisation carries the risk that you might lose focus on the overall target because of focusing on the details. The roof on the Lean house symbolises that everything you improve must benefit the customer and the organisation.

What is Six Sigma?

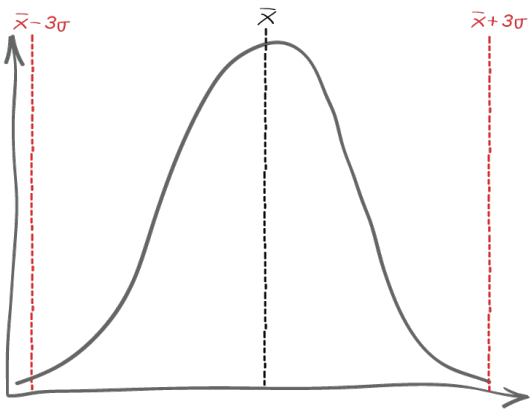
Six Sigma was invented at Motorola at the start of the 1980s when engineers there began to measure quality based on the number of defects per million opportunities (DPMO). Using this as their starting baseline, Motorola engineers developed a method of conducting sys-

tematic quality improvements. Six Sigma became known to the wider public when Jack Welch applied it comprehensively in his overhaul of General Electric at the beginning of the 1990s. Today, it is used by many different industries in both the production and provision of services.

Six Sigma is a method used to identify quality improvements and clarify how existing processes in an organisation can be optimised. Its target is to eliminate defects and reduce variation. For example, by avoiding ordering errors or reducing the average time variation in processing a customer request and thereby being able to plan the number of persons needed at any one time to maintain the established level of customer service.

The little Greek symbol

In statistics, the Greek letter small sigma (σ), is the symbol for standard deviation, which is a measure of how much data measurement points vary around a sample mean. Therefore, if your measurements are widely spread, they will display a larger standard deviation.



Normal distribution with red lines indicating three standard deviations (3σ) on either side of the sample mean ( ).

According to the Sigma Six theory, a high standard deviation is seen in the above figure as an indication of a lack of control over the process. Therefore, aim to identify sources of variation in order to systematically remove them. Six Sigma uses a scale to show how good an organisation is at avoiding process errors. The fewer mistakes a process makes, the higher it reaches on the scale. If a process is at 1.5 sigma, this means that there are 500,000 defects per million opportunities. If the process reaches 6 sigma, it means that there are 3.4 defects per million opportunities. The name 'Six Sigma' was chosen because the quality target of Motorola was to achieve 6 sigma. It is not always relevant to pursue this level, however, as it typically demands a substantial investment of time and money. There is a difference in how precise processes need to be. If an organisation works with essential medicines it will most likely have a lower threshold for mistakes than a company that bakes cupcakes.

The advantage of the Six Sigma scale is that you have a tool to compare processes, for example, before and after improvements are implemented. Moreover, it can be used as a requirement in collaborations with external partners. The Danish wind energy company Vestas, for example, requires all those who wish to supply parts to it to document that their processes reached a minimum of 4 sigma.

The method contributes by using data and statistical analysis to improve performance, existing systems and processes, and not least to ensure that new processes, via on-going checks, continue to function as intended.

To a large degree Six Sigma is about quantifying various factors, i.e. to determine the precise extent of errors and variation, and identify which factors cause these errors and to what extent the individual factors contribute.

The starting point for Six Sigma is a detailed business case that sets a financial target for the project. At its conclusion, the financial outcome is documented — reinforcing the focus on documenting results throughout the project.

Due to the need for extensive training and statistical skills, Six Sigma has mainly been applied by larger organisations which have the resources to build a hierarchy of experts and train large groups of employees in the method. It can be used in smaller organisations, however, when the necessary skills are available.

The background for using both Lean and Six Sigma

There are many examples of large international firms that have generated significant savings by applying either Lean or Six Sigma. However, there are limitations to using only one of these methods. The primary focus of Six Sigma is raising quality and eliminating defects in the process but it does not create flow. While Lean focuses on rapid flow but does not apply the statistical methods to reduce errors in the process.

The two methods can be used together beneficially because:

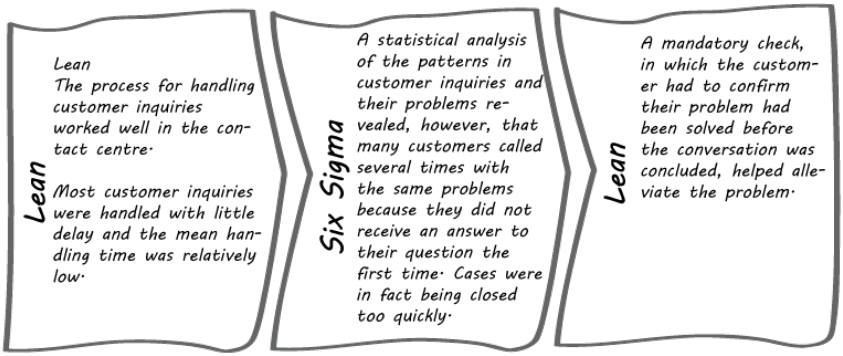
- They supplement each other, creating better results.
- Results are achieved more quickly than if either method was applied alone.
- Resources are put to better use in data collection and analysis due to the complementary nature of the methods.

In addition, Lean and Six Sigma each have some unique advantages which can be incorporated in a Lean–Six Sigma approach:

- Lean: a management philosophy with a complete toolbox and a focus on continuous improvement.
- Six Sigma: data-based, advanced statistical tools, a well-defined project method and project organisation.

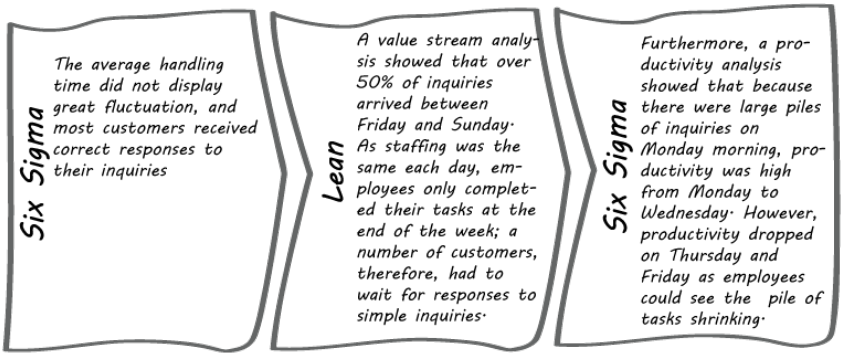
Advantages of combining Lean and Six Sigma

By combining Lean and Six Sigma's methods, a better understanding of process and error can be attained as the following example shows.



Example of combining Lean and Six Sigma in a project.

The problem lent itself to a Six Sigma analysis and Lean was applied in the solution. Below is another example.



Example of combining Lean and Six Sigma in a project.

The Six Sigma analysis did not give rise to any apparent improvement possibilities, the Lean analysis, however, indicated some options which were further highlighted with ongoing Six Sigma analysis.

Even though the methods are applied together, it is clearly possible to focus on one method or the other. There are, however, some fundamental questions that must first be addressed:

- **The maturity of the process**
  If the process is poorly structured or defined, then the starting point should be the Lean toolbox. It is difficult to gather statistical data on an unstructured process. Six Sigma works best when the process is well-defined and universally used.
- **Amount of error and variance**
  If the process generates a large number of errors, then Six Sigma tools should be used. An optimisation of the individual steps in a process that generates errors will not be effective. Therefore, a review of the process using Lean will not solve the problem. By having a data-driven and analytic focus at the start of the project, it is possible to reduce variance. In addition, the impact of a more controlled process will often be reduced errors and variance.

The table below provides a summary of the characteristics of Lean and Six Sigma.

	Lean	Six Sigma
Overall aim	Create faster processes Eliminate waste	Reduce errors in the process Minimise variance
Approach	Process-oriented Observe the work being carried out Involve experienced employees	Mathematically and statistically oriented Gather raw data
Project method	No formal project method	Define Measure Analyse Improve Control (DMAIC) project method
Project selection	Based on value stream analysis with a focus on identifying the greatest source of waste in the process	Based on data and business case(s) with a focus on defining the greatest economic potential
Typical project length	1 week to 3 months	2 to 6 months
Project organisation and training	Apprenticeship, on-the-job training	Formal, standardised training programme with belt colours indicating which level the employee has attained (white, yellow, green or black)

Comparison of Lean and Six Sigma.

Who is responsible for what?

When you are working to improve processes, it is important that everyone in the organisation has the same understanding of their roles and responsibilities. Overall, we see responsibility apportioned in the following way:

Upper management must ensure that sufficient resources are allocated and must champion initiatives and signal that such initiatives are to be taken seriously. Most importantly, they must take ownership of the organisation-wide improvement initiatives.

Middle management must facilitate the ongoing work on improvements, receive input from employees and ensure that improvements are implemented.

Employees must make suggestions for improvements, and be open to change and new ways of working.

When you organise process development as a project, involve upper management, middle management and employees in different ways.

Managers with decision-making capacity and responsibility for the users of the process are involved as the project's steering committee. They must ensure that the project is grounded in the organisation's strategy and that the managerial framework for the project and the solutions is secured. At the beginning of the project, the steering committee must define the project's purpose and target(s), and identify which resources will need to be invested. Subsequently, the steering committee is typically involved in decisions regarding the direction of the project and solutions.

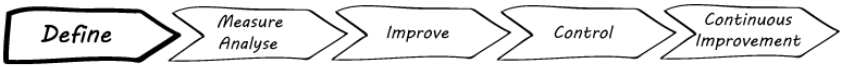
Managers who are more peripheral to the project but whose support is needed are requested to comment on the project's plans, analyses, conclusions, proposed solutions, results, etc. This group will also often participate in management seminars when the solutions arising from the project need to be reinforced at a managerial level.

The process users (those who need the solutions that the project offers), are involved in the *define* phase to identify needs. In the *measure* and *analyse* phase, users are also involved, as they can provide input or data to analyse and can validate interpretations and conclusions. In the *improve* phase, users are involved in providing input for proposed solutions and perhaps also testing and commenting on those solutions. In the *control* phase, solutions are handed over to users. Users may be employees, or upper or middle management.

Representatives of users and management should be involved as well as, on occasion, experts in Lean and Six Sigma solutions, as project participants in defining the project, conducting the analyses, developing solutions, supporting implementation and in the on-going dialogue with the project's stakeholders. Finally, a project manager is appointed who will manage the day-to-day running of the project.

Following this introduction to Lean, Six Sigma and roles and responsibilities relating to them, we will now describe how you can carry out process development projects with the assistance of the DMAIC model.

2 · **DEFINE:** IDENTIFICATION OF THE PROBLEM



Process development projects arise in many ways. In some instances, the organisation has been through an organisational change that requires a change in work processes. Alternately, budgets may be tight and department heads may attempt to optimise an organisation's processes in order to achieve savings. In other cases, the organisation may not be delivering the desired results, or maybe there is a feeling of 'Hey, we can do better than this!'

Some organisations have a fixed procedure for how projects arise, others are more ad hoc. Regardless, the first step should always be to ensure that the task is well defined and that the objectives of the assignment are clear. There must be agreement and support for them, and they must be worth the effort to attain them. A well-defined project has an increased chance of reaching its target and creating value for the organisation, it is easy to direct, and to accumulate support for. These are what the *define* phase must ensure.

This chapter will look at the following questions:

- How do you get started?
- How do you choose what to focus on?
- Who should you involve and how?
- How do you define the targets of your project?

- How do you set the limits of your project?
- How do you plan your project?
- How do you make a business case?
- How do you get quick results?
- How do you summarise, *define* and secure a clear mandate for the future?

These are the central questions that must be answered during the *define* phase. The answers to these questions indicate a way through the phase. To make the process easier to understand, it is described in a linear fashion but, in practice, you will often have to go back and reanswer some questions.

Getting started

Process problems, and their solutions, take many forms. The transition from informal discussions to a formalised project is generally smooth. Begin your run-through when the organisation first begins to work on the task in a structured manner — this is usually at a kick-off meeting. Invite relevant managers and some key employees to this initial meeting. We usually use the following questions as our starting point:

- What is the problem or where does the potential problem lie?
- What would be a good aim for the project?
- What would be proof that the project had fulfilled its aim?
- What must the project deliver?
- Who will the project help?
- Who can affect the success of the project?
- What is our experience of working on similar projects?
- Are there any constraints on the solution or the manner of proceeding?

- When can the project be expected to meet its targets?
- Which of the organisation's resources can the project draw upon?
- Is there a connection between the answers to the above questions?

Organisations can rarely supply complete answers to all of the above questions at the outset of a project. Some answers require a decision, but conflicts of interest and differing opinions can complicate this. Other answers require further information in order to be answered. At the beginning, the most important thing is to find a balance between time spent on detailed analysis and on progressing the work. Therefore, at the beginning of the project, we want to find provisional answers and as it progresses, we will learn more and be able to tie down those answers or adjust initial decisions. The next step in the process is to establish an analytic overview of the project. For this, use the following tools:

- The **Lean house** can be used to make a quick diagnosis of the organisation's process maturity, and where the greatest improvement potential is likely to be found.
- The **stakeholder analysis** addresses the questions: who will use the project's results and who can affect the success of the project?
- The **Supplier, Input, Process, Output and Customer** (**SIPOC**) **method** is used to prepare a preliminary analysis of the processes to be developed.
- The **customer's voice** is used to identify what the customer values and what they would like the result of the processes to be.
- The **target** is a systematic review of the questions: what is the purpose of the project? What indicates that the project has fulfilled its purpose? And what should the project deliver?

- The **milestone plan** outlines the major steps in the project.
- The **business case** outlines the financial rationale of the project: what value does the solution have and what will it cost to achieve?

The path through the analyses may not be linear as they may be interdependent and the conclusion of one analysis may affect the conclusion of another. Carrying out the analyses demands a broad knowledge of the organisation, which is why a stakeholder analysis should be performed early on in the process, as it can provide input on who should participate in the remaining analyses. The conclusions of the analyses are collated in a brief decision document, also known as the project charter, which forms the basis for the subsequent project framework.

Starting the project

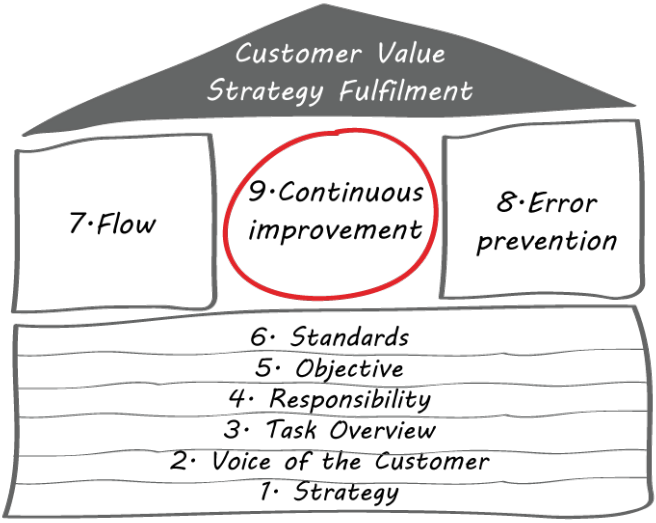
At the beginning of a process development project, the scale of the task can seem daunting. That is why it is important to work on the tasks in a structured way and to quickly identify where best to begin. This will create momentum.

Using the Lean house as a starting point, the project's purpose must be described in order to identify areas which need to be improved. As a general rule, one of the following parameters usually needs to be improved:

- Quality
- Lead time
- Costs
- Motivation (employee or customer satisfaction)

Next, you analyse the organisation's process maturity. Ask questions based on the elements of the Lean house: is there a clear strategy? Has the customer's voice been defined? Is there an overview of tasks? All things being equal, this review will reveal where the problem(s) lie.

The introductory analysis will give you an indication of where it is most appropriate to focus your energies. It is natural to wish to focus immediately on developing flow and eliminating or minimising errors. (This is also where Lean tools come into their own.) However, if the project's foundation is unstable, it is highly likely that any steps taken will be in vain. It makes no sense to optimise flow if each single process sequence is unique. Nor does it make sense to standardise processes if they have no clear target, etc. So the best place to start your initiative is at the lowest level of the Lean house where your organisation is experiencing problems.



The Lean house giving an overview of focus areas and the sequence of improvement initiatives.

The Lean house can be described in many ways but must include an overview of the elements that make up its foundation and supporting pillars. The illustration of the house can be shown to employees so that they have an overview when analysing parts of the processes.

Involving employees

Whether process development is successful or not depends on whether the right employees have been involved at the right time and whether they have a clear understanding of the purpose of their participation. The project's success depends on which employees contribute and what they contribute when designing suitable solutions that both management and employees can commit to implementing. The most important thing, however, is that the decision makers and steering group create the space for the project and its solutions, by securing resources, promoting acceptance and making the project a priority in the organisation. In the final analysis, it will be the project stakeholders who will judge whether the project has succeeded. A project that satisfies stakeholders, while it may have failed on other parameters, can still be considered a success. Stakeholder analysis is vital in order to target work at the project's stakeholders.

How to carry out stakeholder analysis

How to involve employees

Whether you involve employees or not is based on a blunt discussion of pros and cons. The pros are typically that it increases employee motivation, sense of ownership and is enriching. Involvement also ensures dialogue between the strategic and operational levels of the organisation. The cons can be that it is time-consuming and complicates directing the process. Some managers also experience that the involvement of employees limits their leadership options.

We believe, however, that this discussion is often based on the misconception that employee involvement is on an all-or-nothing basis. Discussions on employee involvement should be based on how, when and to what degree they should be involved.

Involvement can happen on several different levels. Our experience is that the degree of involvement should match the decision or task to be resolved and will thus vary within the project. We recommend a high degree of involvement in those tasks and decisions that require dialogue, learning and a sense of ownership among the employees for success. However, this degree of involvement should only be selected for those tasks where management is prepared to allow employees to make decisions and where employees have both the opportunity and desire to assume ownership. For example, it does not make sense to involve employees in questions such as: 'Who should be fired?' In addition, it is important to adapt the degree of involvement to the organisation and its unique culture.

An 'involvement stairs' can refine discussions on involvement

We use the 'involvement stairs' to refine our considerations on employee involvement. They describe the requirements and consequences that the involvement has for the project and suggest activities for the various degrees of involvement.

The purpose of stakeholder analysis is:

- To reveal who affects and who is affected by a project
- To evaluate the stakeholders' significance to the project
- To develop initiatives which answer the demands and expectations of stakeholders

Stakeholder analyses usually take place at kick-off meetings with the project team,

which preferably includes representatives of affected managers and employees.

1. Begin by brainstorming as to who qualifies as stakeholders in relation to the changes you are about to make.

2. Write stakeholders' names on flash cards or Post-its. Spread them out on the table or stick them to a wall to gain an overview. Visualise the lifecycle of the product from beginning to end. Consider the interests of internal as well as external stakeholders. The more precisely you identify stakeholders and their interests, the better your cooperation with them will be. The end target is to name stakeholders or representatives of groups of stakeholders.

3. Prioritise stakeholders — who are the most important and why?

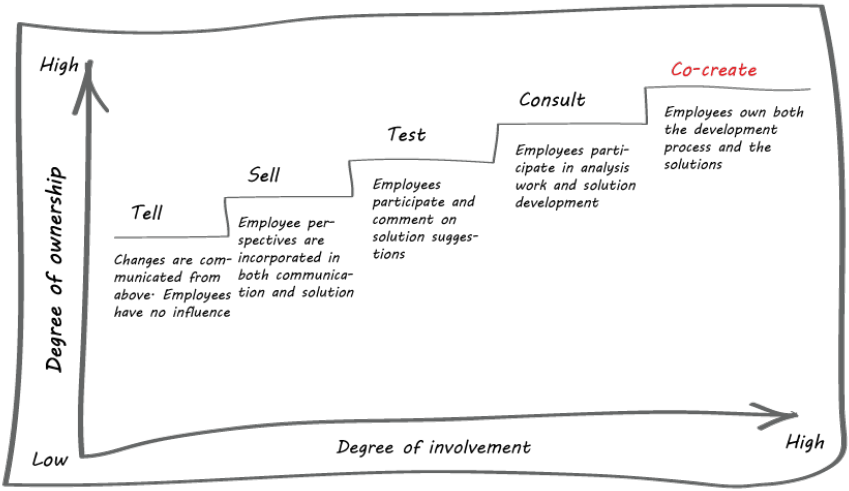
4. Describe the pros and cons of executing the project and the result of the

project from a stakeholder perspective. Describe how you will involve each individual stakeholder. Identify one responsible person and a deadline for this.

5. Find out the names of those stakeholders you do not know. (This may require some research on your part.)

6. Validate the analysis — when possible, consult the stakeholders and discuss their opinions of the project, their assessment of the advantages and disadvantages and whether they are satisfied with how you plan to involve them.
   If possible, ask the stakeholders to identify any other stakeholders which you may not have identified.

7. Set up an archive of stakeholders and how they are involved. Update the archive as the project progresses and as relations with stakeholders evolve.



Outline of the 'involvement stairs' showing the levels of involvement and ownership in a project.

Case 1: early involvement of employees.

When an organisation starts to talk about process optimisation, employees anticipate cutbacks in spending, lay-offs and restructuring. Some will have experienced something similar before and, whether you like it or not, will view any changes with suspicion once they hear processes are being reviewed.

Therefore, it is crucial that you maintain open communication. The first thing you should do is tell people why you are there, what you are going to look at and who employees should go to with any questions or suggestions.

This approach proved decisive in a project we worked on for a public body. One of the first things we did was discuss the project and its goals with the entire department thus mobilising employee support early and allowing us to continue our work

as process consultants. Was this a genius plan? In reality, no. We chose to present our project to employees the day after our introductory meeting as a department meeting was scheduled, which we were invited to attend. We told employees that we were in the initial phases of the project; that our ideas for possible solutions were not fixed but that we wanted to help the department to get schedules and cases under control. We also told them that we were not there to fire anyone but to improve working practices so more could be achieved with their existing time and resources.

A simple picture was used to illustrate that we needed the entire employee — not just their arms and hands but the whole person.

Using their heads, they would be experts we could consult when we needed answers to questions about their daily work or areas of expertise.

Using their hearts, they would be ambassadors, committing wholeheartedly to the project and speaking positively of it with external partners.

Their legs would carry them from place to place when they undertook the hard work involved in the short but sometimes

demanding implementations during which no one should feel above making a photocopy, taking notes, documenting processes, etc.

Due to the openness of our approach from the beginning, we succeeded in establishing significant trust with the employees who felt free to make many useful suggestions and assumed considerable responsibility during the implementation.

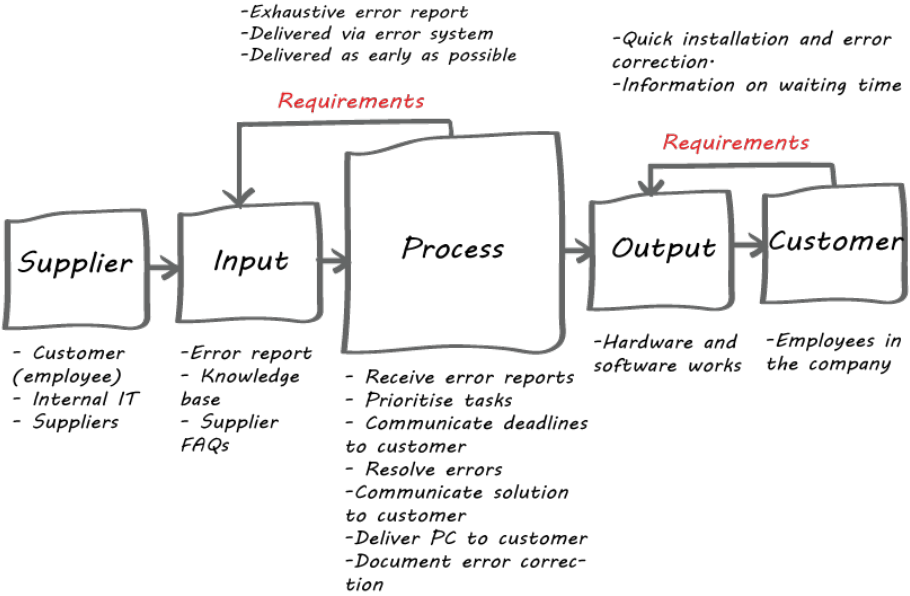
How to create an overview and delimit the processes you need to prepare

In the *define* phase, you create an overview of the processes to be developed; the customers of the processes; the demands of those customers as well as the process dependencies. The initial analysis of the processes must ensure that we have identified the right focal areas for the project. The aim is to have an overview that will allow you to establish rational project limits without spending too much time on analysis. It is a waste of time, effort and money to try to optimise a redundant/irrelevant process. For example, if an organisation's complaints-handling process is slow, it might be best to begin by inquiring whether the number of complaints can be reduced rather than by trying to create the world's best complaints-handling system from scratch. At the same time, problems with the process can be due to problems with input, and these can be difficult to solve by just looking at the process itself.

An important tool for understanding the course(s) of a process is SIPOC as mentioned earlier (chap. 1).

With the help of SIPOC analysis, we can achieve an overview of priorities from a customer perspective, also known as the 'customer's voice', which we spoke of earlier (chap. 1).

Below is an example of SIPOC in use in an internal IT process. Customers had complained about a lack of information regarding repair waiting times. In the 'Input' section, problems are often evident with error reports that are incomplete, or that are submitted outside the official error reporting system or are submitted with a very short deadline. All of these are circumstances that will make process improvement difficult if only process activities are examined.



Example of a Supplier, Input, Process, Output and Customer (SIPOC) diagram providing an overview of the process and its input and output.

With SIPOC, you will gain input on problems you can choose to analyse as a priority as well as an overview of what should be covered by the analysis; which products or services are relevant and how much of the process should be encompassed. Note especially the demands which run from customer to output and from process to input. By

knowing these demands, you can quickly discover what is critical to the process.

How to carry out a Supplier, Input, Process, Output and Customer analysis

Before you begin to fill out a SIPOC analysis, it is crucial that you define precisely which process is to be analysed, and where it begins and ends. If there is disagreement on this, it may prove difficult to reach agreement in other areas later on in the project. The easiest way to fill in the model is from right to left (i.e. your starting point is the customer and your end point is the supplier).

Using the customer's voice

The SIPOC diagram gives a general picture of the customer's attitude in relation to the output of the process, but now and again it is important to be more specific in your understanding of the customer's attitudes. The way we do this in Lean and Six Sigma is by listening to the *customer's voice.*

In Lean, the customer's voice is used to define value-creating elements in the process and subsequently to identify waste. Everything that does not directly contribute to creating value for the customer is considered waste. In Six Sigma, the customer's voice is used to set the limit of what is an acceptable output of the process: how large can the variation be? How early or late will the customer accept delivery of a product or service, etc.?

We use both approaches to the customer's voice in our process work. Additionally, we use the customer's voice to determine the objective of the project: what must be done to satisfy the customer? What is the purpose of the process? There is nothing more wasteful than perfecting a process that delivers something effectively that the customer does not need.

The most important principle concerning the customer's voice is that it must come from customers themselves. In almost every organisation there are employees at every level who are convinced that they know the needs of the customer without ever having asked them. A typical assumption is that 'everyone thinks the way we do'. But customers, like people everywhere, are varied and their likes and dislikes can often surprise even those with many years' experience with the process.

Working with the customer's voice includes gathering data on customer needs, analysis of different segments, quantification of customer needs and prioritising different needs. Of course not all needs are equally important to the customer. Some needs will be requirements (need to have), while others may be wishes (nice to have).

When customers express a need, it often takes a subjective form:

'Your organisation never answers the phone when I call.'

'I have to wait months to get an appointment.'

'The person I talk to always sends me on to someone else.'

Customers express their frustration but in order to make use of their complaints, it is often necessary to make them more precise. In Six Sigma, the customer's voice is translated to Critical Customer Requirements (CCR).

In the example below, we show how a subjectively expressed customer communication is translated into a concrete target.

The customer's statement: '*It's hard to get through when I contact you for help.*'

This is reinterpreted as a target regarding how long the customer finds it acceptable to wait for assistance. In this example, a waiting time of 120 seconds is deemed acceptable. The translation of the customer's voice to critical customer requirements can be done via customer interviews, the investigation of complaints and data on how long customers wait before hanging up. The most important thing is to get a detailed description of the customer's experience.

Specific customer demands do not always match the organisations way(s) of assessing its own performance. Therefore, it may be necessary to carry out further translation to what is known as Critical to Quality (CtQ). *This is how an organisation measures if it is fulfilling customer expectations.*

Voice of the Customer (VOC)	Critical Customer Requirements	Critical to Quality
It is hard to get through when I contact you for help.	Short waiting time (maximum 60 seconds) for telephone inquiries.	Service level of 80% of calls answered within 60 seconds.
When I talk to an employee, they cannot solve the problem, and pass me on to someone else.	The employee the customer contacts must be able to solve the problem without passing it on.	First Call Resolution rate a minimum of 90%.

Example of how the customer's voice is translated to concrete target points.

When the customer's voice is integrated as part of the organisation's ongoing performance target, there must be an assessment of how much it will cost to fulfil these demands. The customer in this example wants to get in touch with an employee within 120 seconds, and the company has set its target as fulfilling this in 80% of its calls, as it would cost too much to respond to 100% of calls within the target time.

Target-setting gives the project direction

Target-setting is a systematic and easy way of communicating a project's purpose, and targeting its outcomes and deliverables.

The project's purpose is a brief and precise description of why the

project is necessary. Typical purposes for a process development project would be:

- To increase the company's ability to respond quickly to changes in product demand
- To reduce production costs
- To increase customer satisfaction with the product

The purpose of the project is its strategic anchor within the organisation and should always be founded on the strategy of the organisation or department that initiated the project. The purpose is formulated in broad, general terms and is not measurable.

The target outcome is implementing the project purpose. It is a short description of the concrete measurable benefits that the project must show in order to have fulfilled its purpose. Examples of the target outcome for a process development project are:

- 'By 24 February of the following year, at the latest, the average time between a customer submitting an order to that order's delivery will have been reduced by 25%.'
- 'Within a year, the production costs per unit will have been reduced by 5%.'
- 'Next year's annual customer satisfaction survey will show that customer satisfaction has increased by 8%.'

The project's deliverables describe the end products which the project delivers. Typical deliverables in a process optimisation project are:

- The implementation of a new case handling process
- A training course for employees
- The purchase of new IT system

Deliverables are typically the most substantial results of a project, but in process development projects, they are secondary to the target outcome. The decisive factor in this project is not whether a new production installation or retraining of employees has delivered a saving of 25% but that the effect is delivered securely and cost effectively.

Many organisations focus on a project's deliverables — precisely because they are so substantial and easy to relate to. During the *define* phase, examples of project deliverables are listed to help stakeholders to visualise the project. However, it is important to underline that these are only examples. In *define,* the only deliverables that are decided are those that are delivered in the *measure and analyse* phase. The final deliverables are first formulated during the *improve* phase.

The target outcome and deliverables must be clear and well-defined. As a general rule, they must be **Specific**, **Measurable**, **Accepted** and relevant for the most important stakeholders, **Realistic** and **Time-specific** (SMART).

In addition, it is important that the goals as a whole balance the needs of customers, organisations and employees. It is rarely profitable, for example, to reduce employee costs alone by pressuring employees to work harder. This usually gives rise to resistance and is tiring for employees. Similarly, reducing quality is unlikely to be a sustainable solution. Striking a balance between the project goals must ensure that the project is viewed holistically and that problems are not solved in one area only to have others generated elsewhere. The timespan of goals is often also important. Short-term and long-term goals should be considered when formulating the target outcome.

Exact goal-setting can pay off

It is crucial that a project which develops processes has a clear goal. It may seem obvious but we see an astonishing number of projects being carried out without a clearly stated purpose or goal. Moreover,

there are many benefits to spending time setting a project's goal.

A clear goal statement strengthens the direction of the project. A project with a well-defined goal is much easier to direct than one without a goal, which allows project participants to view the project goal in different ways. In the absence of a clear goal, uncertainty may arise as to when the project has been successfully completed. Furthermore, the lack of direction increases the risk that the project might fail.

A clear goal statement strengthens project communication making it is easier to communicate the content and status of the project to shareholders. It also makes the delegation of responsibilities/tasks easier.

A clear goal statement strengthens the quality of the project. Exhaustive work on setting the project goal increases insight into the task at hand and creates a holistic view of the project. We tend to focus on immediate solutions, but time spent on deciding the project aim will often allow the project team and steering committee to choose between alternative solutions and find the best one(s).

A clear goal statement strengthens prioritisation. It is easier to evaluate different projects and prioritise them in relation to each other when you already have a clearly stated aim. Moreover, suggested solutions for individual projects are easier to evaluate by measuring them against goals. The extra time spent during the start-up period on formulating a clearly stated aim will, as a rule, give returns many times over in the form of time and yield.

Operationalise the stated goal

For some projects, the goal is given in advance, for example, because it has been formulated as part of a strategy process or in the course of establishing a project programme. In those cases, the task is to ensure that the stated aim is qualified and made operational. This means ensuring that it is formulated in a way so that it lives up to the SMART

goal rules outlined above, and so that it measures what we would like to achieve. The stated goal must also be relevant. Perhaps the objective has become dated since it was first formulated and would benefit from a rewording. In addition, the aim must be understood and accepted by project participants and stakeholders to ensure a common base for the continued progress of the project work.

How to facilitate dialogue about the stated objectives

If the goal has not already been formulated, this will take place during *define*. In such cases, we often begin by discussing objectives on an overall, general level with the project's steering committee. We then finalise the stated goal with the project team in dialogue with the steering committee and key stakeholders. In all instances, be aware that it may be necessary to adjust the project objectives during the project as solutions are developed, project participants gain new insights or in response to changes in the wider world.

Milestone planning makes your project manageable

At the end of the *define* phase, you'll have created an overview of the project's objective, success criteria and key deliverables. You'll know who the key stakeholders are, and have created an overview of the processes to be optimised. Therefore, it is possible to begin to outline how you can approach the project. In other words, the first version of the project's milestone plan can now be created.

The milestone plan is based on the objective formulation which was broken down earlier and supplemented with the list of deliverables — deliverables that we think can deliver the desired effect.

In the *define* phase, the milestone plan is split into two levels:
1) The main plan that outlines how the project will achieve its overall objectives, and when they will be realised.
2) The plan for the *measure* and *analyse* phase that outlines what we

want to know, and which analyses can provide this information. In addition, targets for project mobilisation within the organisation are set, as well as those activities that will ensure this mobilisation. Finally, the plan includes an overview of which decisions must be taken — and by whom, when and with what input.

In the *define* phase, the solutions have not yet been designed and therefore the details for the *control* and *improve* phases have not been set. However, it is much easier to communicate the project to relevant stakeholders if there are some examples of possible solutions.

Procedure for milestone planning

You prepare the milestone plans in the project group in consultation with the project's steering committee and key stakeholders. It is crucial that the combined project group is involved in the preparation of the main milestone plan. This ensures acceptance, professional quality and understanding of the plan content. Milestone plans for individual initiatives may be prepared by smaller working groups but should always be discussed (and adjusted as appropriate) with the entire project group in order to coordinate with other initiatives.

Once again, we recommend working with visual aids, such as flash cards or Post-it notes. This ensures participation, overview and a high level of energy. The method of progressing is, in principle, simple but the individual steps can be time-consuming. It is worthwhile to be thorough in this process as it creates an overview and a good foundation for the later direction of the project.

How to make a milestone plans

1. Apply the Specific, Measurable, Accepted, Realistic and Time-specific (SMART) principles to the objectives from the objective formulation workshop and align them with the success criteria.
2. Sort the deliverables into project tracks. In other words, group the deliverables with some commonality together and separate those that should be tackled individually.
3. Main milestones are identified for each initiative. These are the most important decision-making points or the most critical partial deliverables. In order to break down the objective statement, ask: 'How can we reach that target?' It is a good idea to formulate milestones in the past tense, for example 'Telephony system implemented'. This helps you to ensure that you don't track activities, but actual milestones completed.
4. Identify the milestones which are dependent on one another to identify a critical path.
5. Plan the sequence of milestones based on the critical path.
6. Apply SMART principles to each of the milestones.
7. Evaluate milestones with key stakeholders. Do the milestone plans fit with theirs? Do the stakeholders involved have the time? Can the organisation allocate the necessary resources when needed to reach the objective according to plan?

The milestone plan is the backbone of the project. It creates overview, structure and systematics. The project manager achieves the required management basis for planning how the targets can be reached. Milestone plans must be actively applied in monitoring the progress of the project. Make sure the plans are visible. For example, enlarge the plans and hang them up in a prominent location in the office. Mark clearly the progress of the project according to the plans and write any changes into the plans.

The milestone plan is an important input for the project's budget and business case. We will look at the latter in the next section.

The business case enables financial prioritising

The business case is part of the project charter. Its primary considerations are economic, but considerations such as employee satisfaction, ethics and legal limitations may also be considered. The purpose of the business case is to make it possible for the organisation's management to allocate the organisation's resources in relation to which interventions should be prioritised and which might be postponed or deselected.

Developing the business case is a continuous process. In the beginning, the business case builds on assumptions and estimates of the value of the project's proposed solutions, and the cost of implementing these. In the course of a single project's life, the business case is continuously updated in order to assess whether its commercial rationale remains realistic. If it turns out that the value the project creates does not justify the investment of the resources required to complete it, then the project is shut down.

How to make the business case

The business case is made by the project manager or the responsible departmental head with input from the project group, steering committee, specialists and key stakeholders.

How to make a business case?

1. Determine if the organisation has carried out similar projects before. If it has, then use them as input.
2. Evaluate the financial value of reaching the projects goals. Since these can be based on assumptions, they may be uncertain, and it is advantageous if you can base the value on data from previous business cases.
3. Draw up a project budget based on the milestone plan. The budget should be based on estimated costs for individual activities; historic data from comparable projects (if available) as well as evaluations of which resources you will invest to achieve the desired result.
4. Add a budget for risk management. If you want to be able to change the direction or content mid-project, it is a good idea to also add a budget line for project alteration.
5. Describe the non-financial costs too, such as stress in the organisation due to change, and the non-financial benefits, such as an improved work environment in the long term.
6. Describe the assumptions which form the basis of the budget and the effect of the success criteria. Assess whether they are realistic.
7. Present the business case to the project group and stakeholders and discuss it with them in order to achieve consensus about the project's assumptions and financial objectives.
8. Agree with the steering committee on the degree of tolerance within the organisation for change in project costs or outcomes.
9. Continuously update the business case as the project progresses, so as to adapt to changing circumstances and new knowledge.

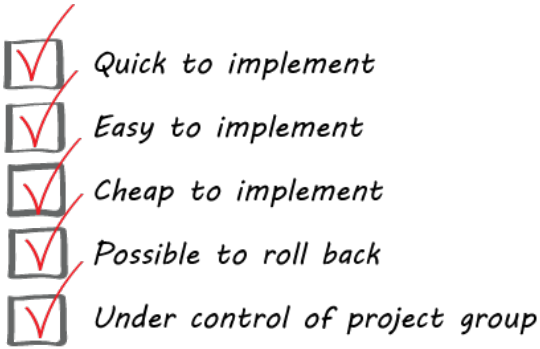
Quick wins

An important element in the conclusion of the *define* phase is the identification of and plan for the implementation of 'quick wins' — i.e. small initiatives that quickly return a positive outcome. Quick wins are important because they often create momentum for, visibility of, support for and confidence in the project. As we mentioned before, every project has a knowledge deficit at the beginning. Implementing quick wins can accelerate the learning curve and provide important information as to how the project's other solutions can be

implemented. In some instances, quick wins alone are sufficient to achieve the desired effect.

An example of a quick win is the disposal of tasks that are no longer relevant, for example, reports that are not being used. It can also be the establishment of simple standards, for example, preparing templates of standard letters to streamline administration, or preparing checklists to minimise errors.

A good quick win is characterised by the elements shown in the figure below.

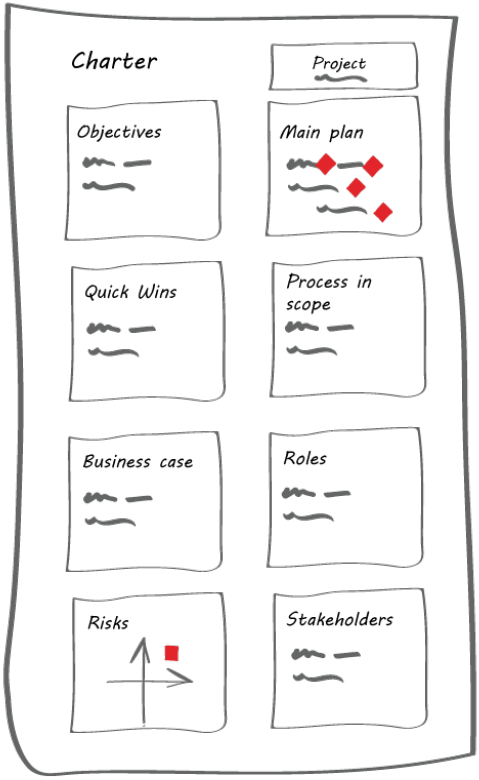


Checklist to identify if improvements are quick wins.

Quick wins can often be identified in conversation with stakeholders and when preparing the milestone plan, but in most cases they need to be fleshed out before they can be implemented. Their implementation must be formally approved by the steering committee. This will often occur at the end of the *define* phase. However, it is important throughout the course of the project to keep an eye out for any new quick wins which might appear.

Decision on the continuation of the project

The preceding analysis has ensured that you have created an analytical overview of your project, and your involvement has created ownership of your conclusions. The *define* phase concludes with a decision on whether the project should be executed or not. This decision is taken by the steering committee on the basis of a brief description of your most important conclusions, called the project charter. Charters are used in larger organisations to ensure that there isn't an overlap between projects and that the timing and sequence of individual



Example of a project charter providing an overview of a project.

projects' initiation and implementation is correct. For you as project manager, the project charter is a tool to help to maintain focus on the target. It also functions as a communication tool which can be usefully shared with project participants.

Every organisation has its own template. Above a series of typical elements that can make up a project charter are shown.

Before formal approval by the steering committee, it is a good idea to provide key stakeholders with the opportunity to comment on the charter, including steering committee members. Remember that the charter should be a concise summary of project ideas.

Summary

In this chapter, we reviewed project work in the introductory phase: *define*. Initially, the purpose of a process development project was defined as *improving the organisation's ability to create value for customers and meet the organisation's strategy.*

We described how to use the Lean house for an initial diagnosis of an organisation's process maturity and to discover where in an organisation's processes it would most likely be appropriate to start work.

We argued that the success of a project is decided first and foremost by involving the right people in the right way. To determine who should be involved *in what, to what extent, how and when,* we used the stakeholder analysis and involvement stairs and separated stakeholders into five different groups.

In order to define limits and create an overview of the processes to focus on, we used SIPOC, the starting point of which is the customer. Next, we described the output and the specific demands that the customer has to the process. We then described the processes that generate the output and finally the input, and the process's demands for input from the supplier.

In Lean and Six Sigma, the customer's voice is a key term that is used to define what is of value and what is waste to the client, and the degree of variation that the customer will accept in a product or service. The most important principle of the customer's voice is that opinions taken into consideration MUST come from the customer and not from assumptions by the organisation's employees. Nevertheless, translating a customer's subjective statements into operational targets is an important task, but, like all planning tasks, it is essential when initiating a results-driven process review.

We emphasised the importance of the formulation of clear objectives with a systematic description of the purpose (why are we engaging in the project?), success criteria (the tangible measurable benefit/outcome which fulfils the project's purpose) and deliverables (tangible output of the project).

Based on the stated aim, a milestone plan is prepared with a main plan that sketches the most important decision points and interim sub deliverables as well as a more detailed plan for the *measure and analyse* phase. This establishes an overview and baseline for the project.

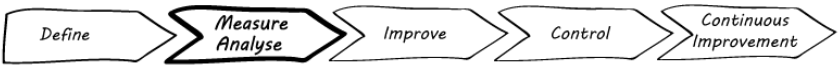
The milestone plan is an important input for the business case, which in turn reveals the project's financial rationale: what benefit can the organisation expect and what investment is required?

In order to ensure progress, visibility and quick impact, it is important to identify quick wins and describe how they could be implemented.

Finally, the conclusions arising from the above work are gathered in a brief project charter with a recommendation on whether the project should be pursued or not. The project charter forms the basis of the steering committee's decision on continuing with the project or not.

With these areas explored, the project is now ready for the next phase: measure and analyse.

3 · MEASURE AND ANALYSE



In this phase, we begin to collect data, map processes and discover where the possibilities for improvement lie. The aim of the phase is for the project group and decision makers in the organisation to reach agreement on relevant issues and make a knowledge-based decision on which processes must be improved and what their improvement potential is. In relation to change management, we use this phase to get employees involved in the project, to create a shared understanding of the issues and gain acceptance of the overall solution framework. As consultants, when we encounter an organisation, we often observe that there is no shared understanding of where in the process the problem(s) lie(s) or what is the extent of those problems. Sometimes the organisation does not even have a clear picture of how the process functions in practice. At other times, management or employees take decisions based on gut feelings and sentiment rather than information and data. Perhaps they overlook the full context of the problem or alternative explanations and solutions, or the fact that the rest of the organisation is not on the same page and is therefore not ready to implement the solution.

In our experience, it is important to create a shared understanding of the problems. After that, it is important to document and communicate the location and scope of the problems. This ensures the quality

of the solution and implementation. The *measure and analyse* phase focuses on that task.

Measure and analyse are often treated as two separate phases. The *measure* phase concludes with a series of well-defined deliverables which lead into the *analyse* phase. The *measure* phase involves data collection and, ideally, this should be performed objectively but more often than not hypotheses are formed simultaneously and may even prove decisive as to whether the correct data is collected or not. That is why we combine the two phases but keep the names: *measure* and *analyse*.

Lean and Six Sigma have differing approaches to the *measure and analyse* phase. Lean does not focus on the mathematical precision of the data. For example, the most important thing is not whether a case is delayed by 50 or 55 days, but that waste is reduced.

Six Sigma focuses on data and precise metrics. The ideal here is to be in a position to translate statistical data analyses of process output(s) to rational decisions. Therefore, preparation of data collection, ensuring the data is well-defined and unambiguous, and validating the results of analysis make up a large part of the phase.

The most important ability in this phase is to ask the right questions: what do we investigate, where do we start? We use the following method in each project:

- Gain an overview of the tasks and problems.
- Prioritise the problems.
- Select analytical tools.
- Carry out data collection and analysis.
- Take ownership of the conclusions.

We will now review a series of the most commonly used tools in Lean and Six Sigma.

Follow the process from A to Z

Gemba is a central concept in Lean theory. In Japanese, *Gemba* means 'the real place', i.e. where it happens, where work is carried out and value is created. *Gemba* allows you to understand a process by walking through it where it is carried out, so that all details can be observed in practice. One way of doing this is to interview people involved in the process and to read the documentation. However, you might miss significant information by doing this. Employees may forget to relate routine details. Others might play down waste or be completely unaware of it.

To get the best understanding of a process, we recommend that you follow it from start to finish. If you are looking at order receipt, for example, you should first talk to the employee who receives purchase orders and then follow the order through sales and inventory. You should follow a case through all its steps so that transfers, waiting times, etc., are included in data collection. In this way, you create an overview of what works well and what doesn't. When a process is slavishly followed, it is a good idea to be aware of the following:

1. Backlogged work and tasks — good indicators of waste and waiting time
2. Standards and whether they are kept — indicates if the process functions optimally or if changes can be made in standards and controls
3. People do things for a reason — ingrained routines may, however, have an effect on the processes' throughput or output.

In this form of data collection, it is essential to ask how the work is done and why it is done in that way. Now is the time to ask the obvious questions as they help to ensure that the process is understood from A to Z.

Mapping tasks

Next, you map the main tasks of the organisation in order to create a quick overview. To limit costs, you will often begin with a single department. The overview will be used throughout your project, for example, to prioritise where to begin and to highlight where time will be spent. Often, it is surprising for employees and management to see where time is spent in various processes.

A loosely structured interview of employees will provide a lot of input with open questions being a good way to reveal the nature of various processes. Such questions include:

- What is the purpose of the task(s)?
- How long do you spend on each task?
- How many tasks do you complete per day?
- How are tasks distributed?
- How are tasks prioritised in busy periods?
- How are cases or items stored between process steps?
- Can everyone perform the task?

Again, we prefer to work visually with flash cards, Post-it notes or a whiteboard; this allows you to show the map of various processes, and their connections, to employees. Remember, the goal here is to get an overview of the total amount of tasks not of how long individual tasks take. This will be addressed later in the process.

The figure below is an example of task-mapping for a law firm.

Every time we have provided participants with a task map, they have always commented on its usefulness. The main aim of interviewing an entire department when task mapping is to clarify the extent of employees' daily tasks and the interdependence of those tasks. Visualisation is also a useful tool for management to get an idea of a

Tasks	Purpose	Time	Volume	Allocation	Prioritisation	Storage	Can everyone?
Sort post	Distribute post	30 min	2 per day	Janice does it	3	Post rack	Only Janice
Obtain the pending	Join cases	10 min	30 per day	Secretary group	2	Secretary rack	Yes!
Print charge sheet	Prepare to post	10 min	40 per day	Secretary group	2	Secretary rack	Yes!
Summon witnesses	Prepare court date	10 min	200 per month	Secretary group	3	Secretary rack	Only secretaries
Send to court	Prepare court date	10 min	200 per month	Secretary group	2	Secretary rack	Only secretaries
Schedule court date	Prepare court date	10 min	200 per month	Secretary group	1	Secretary rack	Only secretaries
Follow-up witnesses	Prepare court date	10 min	200 per month	Secretary group	2	Secretary rack	Only secretaries
Receive verdict	Finishing work	10 min	180 per month	Secretary group	4	Post rack	Only secretaries
File cases	Finishing work	5 min	180 per month	Secretary group	5	Secretary rack	Only secretaries
Handle appeals	Finishing work	10 min	20 per month	Secretary group	2	Secretary rack	Only secretaries

Task summary giving an overview of the most important tasks in a law firm.

typical employee work day. Moreover, the overview will often reveal problems or difficulties with long throughput times.

By also dealing with prioritisation, you can open up this necessary discussion to participants. Lastly, ask if everyone can carry out a task, the answer can indicate what bottlenecks and single points of failure might exist. If Janice, for example, is the only person who knows how to sort the mail correctly, then you may have a problem.

Prioritisation of problems and tasks

Mapping creates a shared overview of the most important tasks in the department. You can see which tasks take the most time and which employees have unique skills. By this stage, you'll have walked

through *Gemba* and talked to the most important stakeholders in *define* and the introductory stage of *measure and analyse*. Generally, at this point, you will have a clear idea of which part of the process is causing problems, as well as which problems are most important to tackle. Now you prioritise the problems that you will analyse more closely.

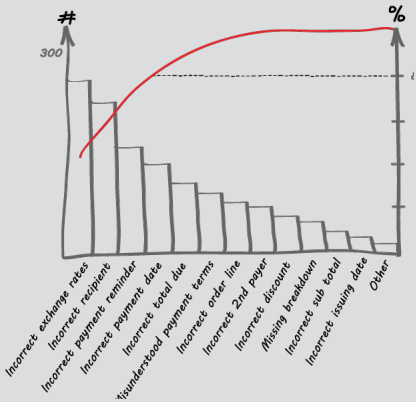
Identifying the need for change will most often begin with a feeling that an area is problematic or could be improved. Next you will carry out a thorough analysis of the problems in the process and impress on all stakeholders the extent of the problems. The path to a solution is rarely simple; you will often have to try a number of different tools and analysis methods. To this end, we often use a 'Pareto chart', named after the Italian economist Vilfredo Pareto. A Pareto chart (also called a distributed histogram) plots the frequency of a given task from frequent to seldom. It also allows you to see each task's share of the cumulative total of all tasks in the data material.

In Six Sigma, this distribution is used to prioritise problem areas as there is a Pareto principle called the 80-20 rule, which states that 80% of problems stem from 20% of the errors in a process. For this reason, the chart contains a line to indicate the cumulative value of each category in order to read the percentage weight more easily.

Case 2: Using a Pareto-chart to prioritise effort

During a project for a major transport company, we noted that one in seven invoices dispatched contained errors and customers were fed up with calling customer service to request revised invoices. So we began to log the types of error that were appearing over a one-month period and found the following:

Error type	Number
Incorrect exchange rates	271
Wrong recipient	238
Incorrect payment reminder	78
Incorrect payment date	50
Incorrect total due	36
Misunderstood payment terms	27
Incorrect order line	21
Incorrect 2nd payer	18
Incorrect discount	14
Missing breakdown/ specification	12
Incorrect subtotal	7
Incorrect issuing date	5
Other	4



Pareto analysis is used to identify the areas with the largest potential.

The Pareto chart above shows which error types most frequently occur in the invoices and therefore the areas which must be included in an analysis of improvement possibilities. When preparing a Pareto chart, various perspectives can be selected. Above we chose to look at the number of inquiries. We might equally have looked at time spent on correcting individual errors or the financial impact of not correcting the errors.

Choose the right tool

Various tools all use the same process map as their starting point. The table below will help you to choose where to begin.

Problem	Tool for analysis
Lacking overview of the process	Process diagram
Unclear roles and responsibilities Poor handover of tasks	Swim lane diagram
A lot of time spent on transport and movement	Spaghetti diagram
Problems with flexibility	Task families
Delivery time too long and processes too expensive	Value stream analysis
Large amount of waste	Waste analysis
Bottleneck problems Delays Unbalanced load on employees and production facilities	Flow analysis
Variation or many errors in output	Six Sigma analysis

Overview of problems and relevant tools for process analysis.

You map the process in a workshop with people who have extensive knowledge of it. This might be the employee who carries out the task or their department manager, etc. The first question to be answered is which process needs to be mapped. More exactly, where does it begin and end? If you review a partial process, you could be excluding relevant problem areas elsewhere in the process. However, if you overestimate the size of a process, you might not have enough time or resources to review it in detail. Consider how much time you have and what the aim is. Which part of the process has an influence on the projects objective? Avoid trying to solve all your problems in one day. Instead, make a list of relevant problems which due to time restrictions will be dealt with at a later date.

If processes are carried out in different ways by different people, note the examples of the differences described. If it is important that va-

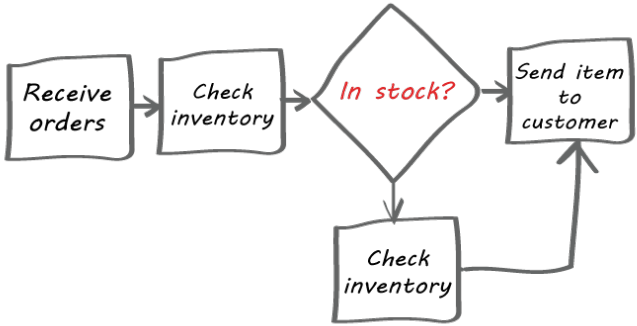
rious departments, etc., take ownership of the analysis results, it is a good idea to involve them in the work.

Flow diagram

The flow diagram is used to create an overview of all the details in a process: what happens when? Which paths does the process follow?

It is often used to gain an overview of what is happening in the process; how the process varies according to product type, issues, output, etc.

A simplified process flow diagram for a process order sequence might look like this:



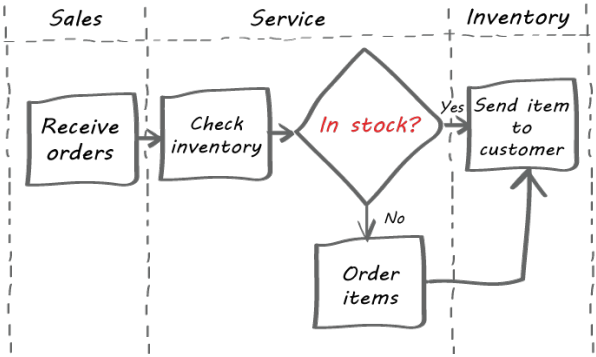
Example of a mapped process in a process flow diagram.

The order is received and an employee checks if the item is in stock. If it is, it can be dispatched to the customer, if it isn't, it must be ordered. This decision is illustrated by a diamond labelled 'In stock?' The individual process steps (rectangles) contain a series of sub processes and tasks not shown in the diagram, which gives an overview. In other instances, you might also include intermediate steps that lead from input in the form of an order to output in the form of goods dispatched. The chart will thus show if there are problematic issues in the process.

Visualisation of the process also helps employees who work on it daily to see their work more objectively and be more solution-oriented when resolving problems.

Swim lane diagram

Use the swim lane diagram when there is a lack of clarity around roles and responsibilities in the process and poor handover of tasks. The swim lane diagram resembles a process chart but swim lanes have been added in the form of dotted lines that show which organisational unit is responsible for each activity.



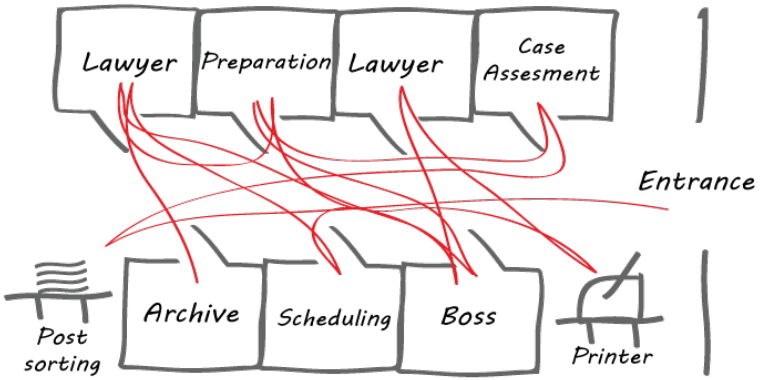
Example of a mapped process in a swim lane diagram.

The idea is to clarify where handovers take place within the process. This is useful to map as handovers often result in waste as misunderstandings/mistakes occur when a task is passed from one person or department to another. Delays can occur when a task becomes last in a pile of tasks in the next link in the chain. It can also increase time spent, transport cost and movement to hand the task on, or extra checks and duplicated work if departments are not coordinated.

Spaghetti diagram

Use the spaghetti diagram if too much time is used on transport and movement in the process. Sketch the physical layout of the workplace and then sketch the movements which happen within the process (shown in red in the figure). The result is often surprising.

Very few people consider how much unnecessary movement there is in their processes because they have become accustomed to the design of the workplace. The spaghetti diagram is subsequently used to redesign the physical layout of the office, workshop or warehouse.



Map of a process, in a law firm, in a spaghetti diagram.

Case 3: a spaghetti diagram redesigns an entire law firm

On our first visit to this law firm in Copenhagen we arrived at the same time as the in-house postman. We noticed that the post for its 45 employees was sorted at the furthest point from the entrance, and the postman's path crossed with secretaries and visitors on his way there.

We began our analysis by reviewing a case from A to Z with

the secretary in charge of sorting the post. We saw how she distributed the mail to desks throughout the office.

Following the analysis, a range of suggestions were put forward by employees to improve the process. One suggestion was to redesign the physical working environment. A pragmatic idea that was chosen, among other reasons, because we had seen first-hand the problems which the office design gave rise to.

In the *analyse* phase, we determined the preferred office layout following a strict 'process layout'. That is to say, the post would be sorted at the entrance after which it would be distributed on two racks. One rack for secretaries and one for lawyers. The lawyers were moved to one side of the corridor and secretaries to the other. (Originally, secretaries were located at the far end of the corridor.) The next step was to reform the process of case visitation. At the beginning of the process, a lawyer and a secretary reviewed the cases one by one and completed any missing information. When complete, they placed each case on a rack just outside their own office. The core process in which lawyers handled cases and secretaries did the final documentation on them had changed. Now there was a shared rack where a colour and a short description of each rack indicated when the case was estimated to be finalised and how it was prioritised. It was easy for both groups to see what the next case was and how far it had come.

Further down the corridor, the case file emerged again when a court hearing needed to be scheduled. The secretary responsible took the first case file from the rack to the left of their door, scheduled a hearing and when it was confirmed, placed the file on the rack to the right of their office door.

As part of the process improvements, we had to remove a wall

and saw a couple of racks in two but it was a pragmatic and effective solution.

The idea behind the redesign was to make the physical movement of the process visible. The new physical layout of the office eases the progress of the case file through the different departments.

The redesign had a surprisingly significant effect. Previously, lawyers had spent up to 70% of their time searching for case files and responding to questions about a case's status. Following the process reform, all employees could check the case register and locate the correct rack without bothering the lawyers.

Task families

If your organisation is having difficulties with staff flexibility and responsibility sharing, then you should work with task families. Establishing task families simplifies processes and increases staffing and responsibility flexibility. You do this by seeing if some tasks can be classed in groups based on comparable process steps. It is not always possible to have identical process steps, but there will be some that resemble each other and can be grouped to create synergies.

We identify task families by comparing process steps based on:

- Individual product and service types
- The functions which these undergo during the process

The simple tool used here is a case matrix, an example from a municipal office is shown below.

Process	Complete application	Check application	Approve application	Send application	Receive item	Notify citizen
Renewal of driving licence	x	x	x	x	x	x
Renewal of passport	x	x	x	x	x	x
Renewal of social security card	x	x		x	x	x

Case matrix for a municipal office; used to reveal similar tasks.

Despite the fact that there are three different processes, the steps in each process are largely the same. As is the norm, different authorities are responsible for issuing driving licences, passports and social security cards. However, by making the processes as similar as possible, we reduce the complexity and make it possible for employees with the same skills to switch between processes, decreasing the problems associated with peak loads.

For long-term employees, it can be challenging to begin thinking in terms of task families. In some cases, there may be strict divisions between departments to overcome, but, given time, employees will get used to the idea of continuity between processes.

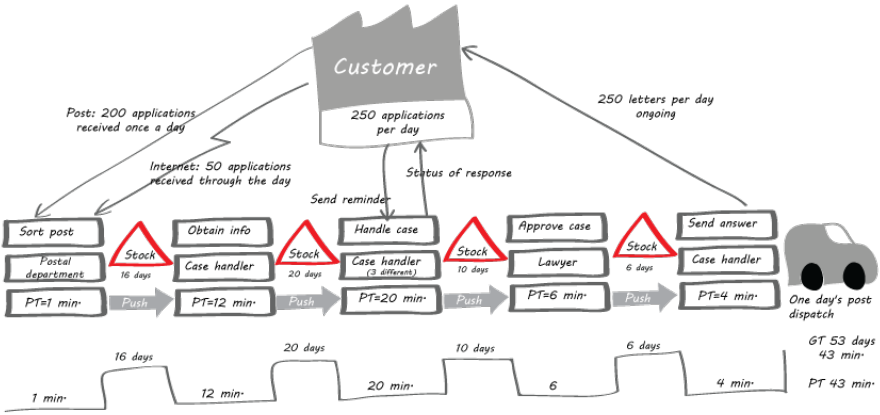
Value stream mapping

If a process is delayed by a long delivery time or is excessively costly, then you should use value stream mapping (VSM). VSM is a core tool of Lean that creates an overview of the entire value stream from the moment the customer's need arises until it has been met.

Unlike a number of other process mapping tools — which begin with tasks, departments and responsibility handover — VSM starts with one single item or case going through the process. The task is followed when an employee is working on it and when it is waiting to start its next stage in the process. Through value stream mapping, you gain:

- An overview of the entire process
- An estimate of how long the case is paused — where does nothing happen in the process?
- An overview of the different types of waste — which process steps do not create value?

Whether the customer is external or internal, it is essential that the mapping is based on their experience of when the process begins and ends. The process ends when the customer feels it has ended and not when the organisation's internal tasks are completed. For example, a VSM will not end when the product is sent to the customer but when the customer has received it and checked its quality. The difference may not be obvious to the organisation but for the customer, the difference will be significant. There may still, however, from their point of view, be defects or errors in the product or service.



Value stream map showing the entire process and those points where the task is being processed, or waiting in a queue or as inventory.

The customer is central as they initiate the process. The two upper rows with rectangles show the process steps and those responsible. The line below illustrates the process and queue times.

If we reduce the value stream further to a single figure, it is the ratio of process time to lead time. In this case, it is 43 minutes (process time) to 53 days and 43 minutes (lead time) corresponding to 0.17% which can be used as the baseline for the project.

Value stream mapping typically includes these elements:

Process steps:

- The general steps a case goes through are mapped, including the time spent on handling the case within that process step.

Activities:

- Each process step consists of one activity or a series of activities which can be carried out in sequence without handovers or queues. Together all activities comprise the tasks which must be completed before the case is delivered to the next process step. (You should note each activity's name along with the average time taken to carry it out.)

Exceptions:

- Exceptions are occasional activities that occur during the process, for example, error correction.

Problems:

- At those points where the process runs into difficulties, notes are taken on the problem(s).

Queues or backlogs

- Queues and certain places in the value stream are where cases or products are stopped thus creating unnecessary waiting time. Problems often arise where there is a backlog and value is lost. That is why it is important to keep an eye on when backlogs occur so you can try to identify the cause. These queues — places where no work is being performed on the case or product — are shown as red triangles.

Process time and lead time:

- At the bottom of the diagram, you can see an overview of process time (PT) — the time used on the product or case — and lead time (LT) — the time from the customer's need arising until it is met. The example above is typical for a VSM: to handle a case, the organisation's process time is 43 minutes but the customer experiences a lead time of 53 days and 43 minutes.

The VSM is always presented in two versions. One, AS-IS, describes the actual current situation and the other, TO-BE, describes an ideal situation or the situation the organisation wants to work towards.

To make the process flow diagrams understandable and applicable across the organisation, it is a good idea to use shared symbols and notation. If your organisation does not have a procedure for mapping processes, you should take this opportunity to establish a shared standard.

Communicate the results of the analysis to the relevant employees preferably through those who participated in the workshop. This widens understanding of the analysis results and enhances ownership among participants, which will be useful in the following stages, including implementation. The more ambassadors the project has, the better. This is especially the case when those ambassadors are the employees who will be working with the processes.

Case 4: 39 people for a full day of value stream mapping

A wind turbine manufacturer decided to map its value stream to generate new ideas. He wondered why, on average, it took 200 days to implement new ideas in production.

Any optimisation of this value stream had a significant impact on the company's bottom line, since most ideas either referred to cost savings (e.g. by using other materials) or to proposals to increase the turbine's power creation or stability.

The process involved a number of departments: Master Data, Purchasing, Manufacturing, Services, Testing, Certification and Engineering departments both tower, rotor blades and nacelles. However, we succeeded in reducing the number of participants to less than 40 for a whole day workshop, where we mapped the process as it then was.

The workshop's aim was to involve those who worked with the process on a daily basis and who needed to agree with the initiatives to be implemented. We could have mapped the entire value stream by asking each employee what they did and sketching this on a diagram. However, this wastes time in the long run because employees must subsequently verify any uncertainties and clarify any disagreements about how things are carried out.

During the workshop, it became apparent that participants had no idea what happened in other departments or how complex the process was. The 'burning platform' was there — change was necessary for survival. This was highlighted by the recognition of the complexity of the process, which dominated the first few hours of the workshop.

A simple way to make a VSM is the 'brown paper method' where sheets of brown paper are hung on a wall. Each process step and activity is then attached to the paper using flash cards

or Post-it notes. At a value stream workshop two weeks later, this mapping is shown to the same participants who can make suggestions about how the new process should be structured.

Waste analysis

If your organisation experiences extensive waste that you want to identify, you should use a waste analysis. By optimising processes, some organisations choose to push employees and machinery to perform better in the hope that productivity will increase. However, there are limits to how quickly people can work and this can result in conflict with employees. The core of Lean is an understanding of what value means for the customer. Those activities which deliver what the customer wants are perceived as valuable. Everything else is perceived as waste.

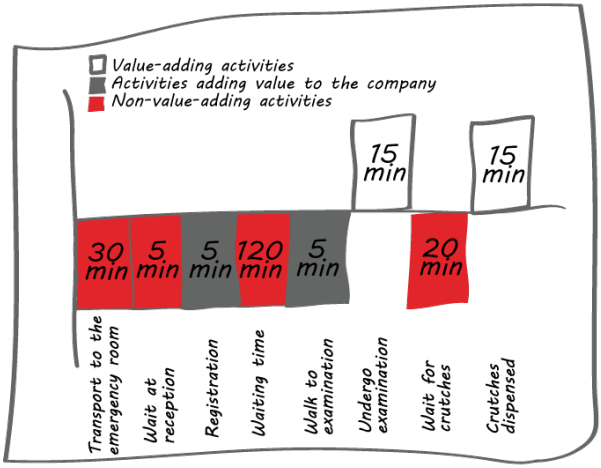
By focusing on eliminating waste, you eliminate activities which typically frustrate employees, hold no value for customers and are costly to the organisation. The potential to identify waste is often large because, as a rule, waste comprises over half of resource consumption in a process.

The first step in the waste reduction process is to agree on which activities have value for the customer. It is not always easy to achieve this. Employees who have worked with a process for a long time may find it difficult to see any of the stages as unnecessary. After all, they will have spent a lot of time working on the process over the years — work which if thought of by the customer as unnecessary will have had no value. It is important to explain the concept well to employees so they will be able to see the distinction between value and lack of value.

A simple tool to facilitate a discussion on value is the overview of activity time against the value it creates for the customer and organisation, respectively. In the analysis, the process is divided into three types:

- Value-adding activities
- Activities adding value to the organisation
- Non-value-adding activities

In the example shown below, we have illustrated the process of accident and emergency visits. The patient wants to get examined by a doctor and receive treatment — in this case, crutches and bandaging. The activities that have value for the customer are placed above the line with all other activities placed below the line. You will note, however, that transport of the customer to the hospital is not included in the diagram as adding value for the customer. This is a good example of a 'necessary' task. Transport will always be necessary due to the hospital's location. This does not mean, however, that the transportation itself has value for the customer. A person in pain would be happier to be treated on the spot rather than being transported to the hospital.



Time and value overview shows where in the process value is created.

Some of the non-value-adding activities for the customer can have value for the organisation because they facilitate how the organisation works. In the above example, registering a patient is necessary to help staff to keep track of patients, while it holds no immediate value for the patients themselves. This is an example of a necessary but non-value-adding task.

Eight types of waste

When detailing waste, there are eight types of waste which occur most frequently:

Waste type	Problem	Example from production company	Example from service company
Overproduction The organisation produces more than is needed or production takes place too early	Ties up capital, increases inventory	Too many products or components are produced because there is spare capacity or because there is a lengthy transition time between production of different products	Too many resources are wasted delivering products to customers earlier than they request them
Over handling/treatment The organisation's performance exceeds what the customer requires	More resources than necessary are used	The product is painted even though the customer does not care about the colour	The customer receives more information than they need
Inventory Time that is not used working on products or services	The customer waits, capital is tied up, products become obsolete; there is no flow in stock inventory, increasing storage costs	Raw material, semi-finished or finished goods remain in stock because of, for example, poor planning or insufficient faith in the processes	A task is halted awaiting the next stage due to, for example, a case that needs to undergo several reviews
Waiting time Time in which the employee waits to carry out their work	The employee is not productive	An electrician waits for a plumber to finish in order to continue with a repair due to distinct boundaries between trades permits	The employee who waits for assistance from a colleague or on a slow IT system
Transportation Moving products, papers or information	Transportation does not add customer value	Items moved from remote storage to storage close by and then moved around several times	Papers sent around between offices
Movement The movement of employees who walk around or otherwise move themselves in connection with work	Time use, potential work injuries	The employee must move from the work area to the store area to fetch necessary tools and parts	The employee needs to click through various screens to find information in the IT system
Error/defects Products that do not work, are delivered late or are lacking information	Additional resources spent on repairs, delays; difficult to foresee	The product is defective	The customer receives an incorrect bill or unintelligible information
Unused skills When the skills of employees are not exploited optimally, they carry out tasks that could be handled by less qualified employees	High wage and production costs	The qualified production engineer is responsible for cleaning up after production, delaying the launch of the next production process	The manager uses a significant portion of his time on casework instead of directing employees

Most common types of waste in organisations.

When identifying waste types, it is important that those involved do not need to decide whether the waste type can be eliminated or if it is sufficiently significant. In this introductory phase, the task is to identify as much waste as possible. Later on, you will decide what can be changed and when. For example, it may seem pointless to characterise transportation as waste in a purchasing process because the goods need to be received by the customer. However, the transportation costs are still waste and when analysing the options, it may be shown that costs can be reduced through better planning, purchasing from suppliers located nearby, etc.

Flow analysis

Work on flow analysis if your organisation experiences difficulties with bottlenecks, delays, unbalanced workloads of employees and production equipment.

Flow is defined as a 'continuous motion of one object at a time through a series of process steps'. In an ideal flow-optimised process, there is no waiting time between process steps. That is, the recipient of a case is ready to work on it, while the person who was responsible for the previous process steps has completed their tasks. When flow works optimally, there is minimal waiting time and minimal inventory.

We would like to emphasise that this is an ideal situation — in reality, there are very few processes where it is possible to achieve the ideal. It will often be necessary to accept waiting times and an accumulation of inventory because of uncontrolled variations in demand and handling time.

Flow rests on the four following central concepts:

- Batch size
- Changeover time or single-minute exchange of die (SMED)
- Variation (*Mura*)
- Production levelling (*Heijunka*)

Batch size

In lean we believe that the ideal batch size is single piece flow, as discussed in the introduction. However, it is rarely optimal to work with batch sizes of one.

Let us first consider why companies often produce products or process cases in a series — a number of units produced at once. These units are assembled later in the process. The series can be based on time or number. For example, there may be two machine hours available to produce as many units as possible. This will give a series of *x* products with *x* being variable. Alternatively, a series can be produced according to quantity where 22 units must be produced and shipped in assembled form. Here, time is the variable and can be changed as part of process improvement.

The concept of batch size is used differently in administrative processes as it is primarily aimed at manufacturing companies. However, the principle can be applied elsewhere. For example, a customer service employee may have only one hour daily to answer email queries because phone calls are given higher priority. Therefore, the number of emails answered depends on their complexity. Conversely, all emails are answered if it is determined that they must be replied to on the same day. The time spent will therefore vary from day to day.

There are several reasons for production taking place in a series. The most common reason is to reduce the time machinery stands idle either by reducing the changeover time or maximising the time spent on production set-up. The longer the machine is available or set up for production of a fixed type of unit, the more such units can be produced. Moreover, if you produce a smaller series, you will spend more time on process changeover but there are other reasons to avoid small batch sizes. The fear of making mistakes during rapid changeovers is another reason. In case work, many people also intuitively choose to work with a series. If, for example, an employee handles a series of different tasks in a week, these should be grouped according to type.

The employee can then write reports on Monday, compile invoices on Tuesday, etc.

The fear of making mistakes in a large series persuaded Toyota, among others, to alter its production plans. It did not want to risk producing cars that no one would buy. Other car manufacturers, however, consider mass production and therefore larger series to be the way forward. Production managers at General Motors and Chrysler — which employ mass production — report that they reach their production plans. Huge quantities are produced every day, therefore, it must be a success. Unfortunately, this is an example of time optimisation where the same car is produced in a shorter time. Both General Motors and Chrysler have made losses year on year, so perhaps they could identify errors in the value chain and optimise their batch sizes.

Changeover time or single-minute exchange of die

Changeover time is defined as the time it takes to change the entire production system to production of a new product. This is often called SMED, which is the reduction of the changeover of production to a single minute. The goal of a single minute is worked towards as an ideal, rather than being a specific target.

Working with SMED is important because it provides the following:

- Better delivery in relation to planning
- Reduced waste
- Greater flexibility

SMED encompasses many different activities, including:

- Filming the changeover to analyse where time is spent
- Developing standardised tools

- Production planning — some products are faster to switch between than others, etc.

In administration, changeover is often the time it takes to get started on a new case or project. The time it takes to switch context is waste and should be reduced. There is a connection between the number of tasks an employee is allocated and the effectiveness of the employee. This must therefore be taken into account in the allocation of tasks. To illustrate the effect of focusing on changeover time, consider the following fictitious example of a bakery's production process.

The baker has several challenges. Currently, he makes one type of pastry at a time and often experiences great waste — sometimes up to half of a series is unsold. In addition, he makes too many types of pastries/breads to fit in the shop; only a few types can be displayed, which makes the bakery's selection seem limited. Lastly, the buns baked in the morning dry out quickly and are sold at half price in the afternoon. The production process for a typical day is sketched below.

Baking process with a lengthy changeover time.

If the baker succeeds in reducing the changeover time from 30 to 15 minutes, he can make another batch of buns as, in total, he will free up 60 minutes per day.

Baking process with a reduced changeover time.

As the baker can now make fresh buns later in the day, he does not need to reduce prices in the afternoon. However, there is even greater waste as he is producing even more buns. In addition, there are several types of bread which are often sold out earlier in the day, with others remaining unsold and being thrown out.

As the baker's changeovers have become faster, he can reduce batch size and constantly evaluate if he needs to make another batch, depending on the demand on any particular day.

Baking process with reduced series size.

Therefore, the baker has optimised the process on all success parameters as follows:

- Quality has increased, as his products do not dry out and can be sold at a better price.
- Lead time is reduced as the production time has been reduced.
- Costs are reduced because the baker no longer discards his products and costly raw materials.

Variation

The third factor that influences the flow in our process is variation. This may be the variation in customer demand, but it can also be variation in the time each process step takes. If there is great variation in production time, the probability is that the product will not be delivered on time. Conversely, in the case of very little variation, the probability is that the product will be delivered on time. In the case of a process with many steps, uncertainty increases since uncertainty

accumulates through the process. Therefore, variation must be reduced in all possible process steps.

Variation will be treated separately later in this chapter in the sections on Six Sigma, the basic philosophy of which is to eliminate variation in processes.

Production levelling

One way of creating flow is to introduce an assembly line production, which runs at the same tempo all the time. In this way, a company can almost force flow into a process. Alternatively, you could introduce fixed-time intervals corresponding to the longest process step. The process step is carried out in the fixed-time interval and the product is sent on as soon as the interval has ended. The principle of the fixed-time interval is called a pulse line. The process gains a rhythm that synchronises production and demand; when the process's products change hands, it is done in 'takt' (derived from the German *Takt* for a conductor's baton). That is, all process steps hand over and receive their product at the same time. The time an interval takes is called the takt time. It shows what the speed needs to be at each process step to achieve flow in the production.

Takt time is set as the available production time divided by the frequency of demand and thus output matches demand frequency.

In manufacturing environments, pulse and takt are central to operation and there are constant efforts to improve flow by changing the takt or levelling the time used on individual process steps. Ideally, each process step is equally long as this will ensure no waiting time between steps. The levelling is done by moving some activities from one process step to another.

Levelling can also be done by adding more resources to a process step. In this way, the process time is halved, and the time gained can

be used to maintain the pace. Alternatively, you could look at the bottlenecks between the process steps. The bottlenecks may be stacks of papers or a build-up of tasks for the employee, who cannot execute them within the desired takt time. If there is a backlog between two process steps, they are by definition out of takt —production is happening quicker somewhere in the process than demand is somewhere else. For example, this may be purchase orders that end up in a pile at the warehouse because staff are too busy packing orders to check those that are incoming.

One of the reasons it can be worthwhile establishing flow in your process is that it lowers the overall process lead time. The example below is one of simple levelling of the process. The time used in each process step is measured and variation in the lead time is shown. By moving activities which take two minutes from step 2 to step 3 a uniformity of time for the three process steps is achieved. This means that a new case can now be handled every three minutes rather than every five minutes. In addition, the inventory that had previously built up between steps 1 and 2 diminishes. Note, however, that the combined production time of nine minutes is unchanged.

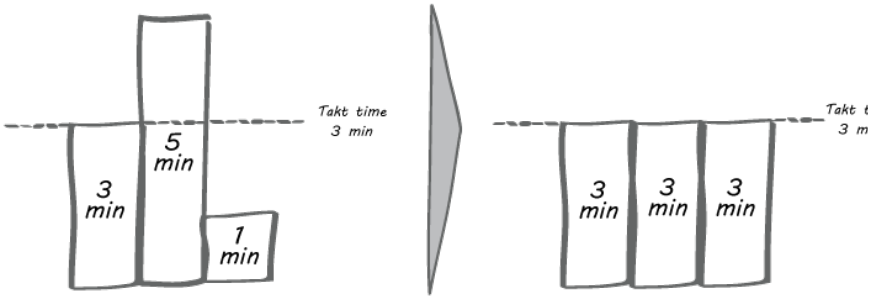


Diagram indicating how flow can be created by levelling a process in relation to takt time.

If your process contains too many complex cases which impact the process time significantly, it may not be possible to create flow. Here, you should consider separating the cases or products according to whether they are difficult or easy to handle. This recognises that some cases or products take longer to carry out than others and must therefore be treated separately. They can be delegated to the most skilled employees or isolated in some other way.

Case 5: an office supplies company separates simple cases from complex ones to achieve flow

At an office supplies company with nine full-time pickers and day-to-day delivery, the delivery van would arrive between 12 and 4 pm to collect the day's orders. Moreover, employees only discovered how many packages the van could take when it arrived at the gate.

Management were confused because they did not know how much work could be performed in any one day. Employees were stressed because they had no idea how much time they had to prepare each order.

When the orders were picked, employees chose the 10 most popular ones and drove around the warehouse collecting the goods to be packed. At the table where the orders were gathered, someone else sorted the goods and packed them. Variation between packages was great, but 80% of them were small orders such as printer paper. The other 20% varied greatly. Some took up to 30 minutes to gather, either because items were in another warehouse or because the packaging itself was time-consuming. There was no flow therefore in all the processes and that created problems and bottlenecks.

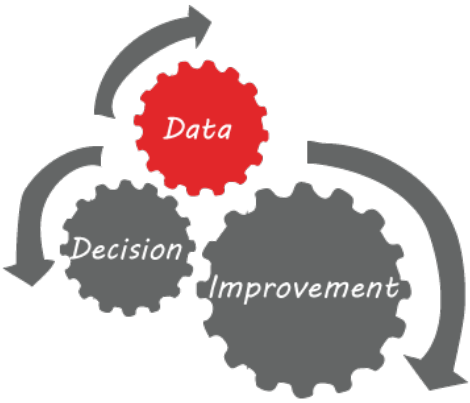
The solution was to renegotiate the route with the delivery

company so that the warehouse became the final stop. In addition, the manager introduced the practice of phoning the delivery van driver daily to inform them how many orders were to be dispatched that day. If there was not enough room in the van, the manager had time to make other arrangements, for example, prioritising urgent cases.

Over and above this, the process was split in two. The most experienced employees were given responsibility for the more complicated cases (20%), while the less experienced focused on the standard orders (80%).

Data collection and Six Sigma analysis

In Six Sigma, statistical analyses of data are used to optimise process output. Data should help us to make the right decisions and ultimately result in improvement. In Six Sigma, the factor to be optimised is the Y factor, while those which affect output are called X factors.



Improvements and decisions are best made on the basis of data.

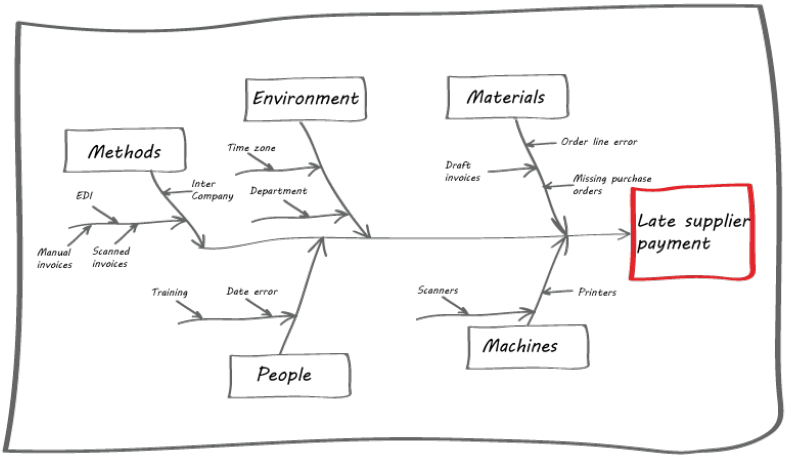
The Y factor is typically identified in the *define* phase during the setting of focus areas and aims. The task in the *measure* and *analyse* phase is to identify the most important X factors and describe how they affect the Y factor. For example, the Y factor may be the number of errors in the process, and the X factors may be employee skill level, machine type, day of the week, raw materials and so on.

The following points must be looked at in order to reveal the X factors and to treat them:

1. Identify potential X factors.
2. Prioritise X factors.
3. Define data.
4. Plan data collection.
5. Run pilot collection.
6. Collect data.
7. Plot and analyse data.
8. Conclude.

Identify potential X factors

The first item in the data collection process is to identify potential X factors that can influence the process performance — i.e. which X factors affect the Y factor. To do this, you should gather the most important stakeholders for a brainstorming session. In brainstorming, we often use a fishbone diagram (Ishikawa diagram). Below is a selection from a fishbone diagram for a Danish transportation company that had problems paying its invoices on time. The Y factor is therefore late supplier payment, and potential X factors are listed under the five headings: machines, people, methods, environment and materials.



Fishbone diagram for a Danish transportation company. This type of diagram is used to brainstorm potential X factors.

Prioritise X factors with a cause and effect matrix

When you have created a fishbone diagram for the Y factors that you want to affect, you will have a long series of potential X factors which can influence them. At this stage, you need to evaluate which X factors affect the Y factors most. You rank the X factors using a cause and effect matrix. In this matrix, X factors are compared based on their influence on Y factors. In addition, the importance of the affected Y factors needs to be evaluated.

Set all the Y factors as the column headings. You indicate their importance to the project in the row underneath using the scale of 0 to 10 — with 10 being most important and 0 not important. Next, add the X factors as row headings and using the same scale indicate how much they affect the Y factors in the column headings. This results in a value in each field and a calculation may be made of the total effect. The value of each X factor is then multiplied by the corresponding Y factor's value. In this example, there are four calculations for each

	Quality	Lead time	Costs	Motivation	
Importance ↓	3	9	0	1	
Input (X-factor) →	Correlation between Input and Output ←				Total
X^1 (Department)	1	3	9	0	30
X_2 (Order line error)	3	9	0	1	91
X_3 (Missing purchase order)	3	9	1	0	90
X_4 (Use of scanner)	9	3	0	1	55

Cause and effect matrix for a Danish transportation company. This type of matrix is used to prioritise identified X factors.

X factor as there are four Y factors, which are summed up at the far right. The calculation for the first X factor is as follows:

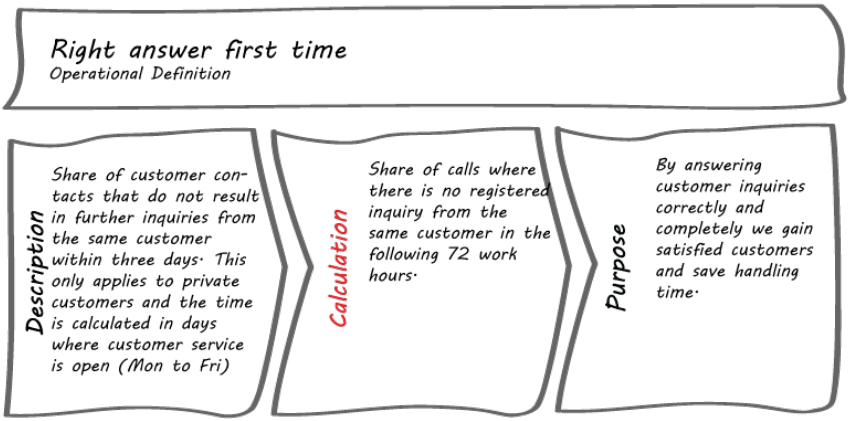
$$X_1: (3 \times 1) + (9 \times 3) + (0 \times 9) + (1 \times 0) = 30$$

In all likelihood, some X factors will have no effect on some Y factors. If this is the case, use 0 so a clear picture of prioritisation can emerge. Afterwards, you can make a prioritised list of the connections between X and Y factors.

Operational Definition

Before collecting data, it is important that everyone agrees on and has the same view of what needs to be measured. This creates the foundation for the consistency and credibility of data collection.

Below is an example of an operational definition in a customer service project focused on solving customer issues on first contact.



Operational definition in a customer service project focused on solving customer issues on first contact. The operational definition is used to ensure a shared understanding of data collection.

An imprecise operational definition can create noise in the data collected as it creates a risk of measurement errors or of overlooking other X factors.

Plan data collection

You can choose to use existing data — as a rule, supplied by an IT system — or collect new data. It can be hard to determine which period to collect data for, especially with a system sample. If the period is short, you risk collecting an unrepresentative data set or one affected by one-off events. If you choose a longer period, you risk collecting data that no longer reflects the current situation due to changes in the organisation, products, procedures and other factors reflecting customer demand, new internal routines, etc.

Finally, it is important to secure the cooperation of all relevant stakeholders for data collection, validation and analysis. It can be a good idea to review the collection with all relevant groups before you begin in order to have all relevant input highlighted.

Run a pilot data collection

It can be hard to foresee every eventuality when planning data collection. Therefore, it is always a good idea to run a pilot data collection. The pilot run should be carried out as realistically as possible — i.e. with the same people, time scheduling, case types, etc. However, run it over a short period (e.g. a few hours) before evaluating the results. See if any surprises crop up or if there are any other reasons to alter your data collection plan.

Case 6: pilot collection cancels all further action

For a Danish shipping company with branches in Central America, Africa and Asia, scanning of supplier invoices was undertaken locally by its Accounts Department. After scanning, each invoice was given an optical character recognition stamp and was uploaded to a global SAP system. The invoice was assigned a document ID and a timestamp. Then the payment process started. The invoices of external suppliers were handled by a global payment centre in Costa Rica and internal invoices were settled locally.

After some time, it became apparent that it took significantly longer to process invoices in Costa Rica than it did locally. A process team was established to investigate the problem. It was revealed, during the initial pilot run of the data collection, that several of the payments processed locally had a negative cycle time. That is, the invoices were registered as paid in the SAP system before they had even been received and scanned. The reason for this was that the times registered for scanning and optical character recognition were based on the *server's time zone* and not the local time, which would have been more accurate.

The solution was to change the calculation of the process time for local invoices. No other changes were necessary as the lead

time at the local offices was correct and therefore more comparable with the lead time at the centre in Costa Rica.

Collect data

Random sampling allows you to investigate a few processes and extrapolate from them. There may be, however, variations that a random sample review cannot reveal. Nevertheless, random sampling is often the only means of data collection as reviewing all the data in a process would be too costly and time-consuming.

In relation to random samples, it is important to consider the following:

- Is the sample representative of the process or data group under investigation?
- Is the process stable?
- Is the data selection random?
- Is there equal probability for selecting any data point?

For practical reasons, you should also consider the cost of random sampling and any obstacles and confidentialities that might influence the ease or otherwise of data collection. If you want to reduce the risk of making an incorrect decision, you should take several random samples. If this data set displays a large degree of variation, you will have to take more random samples.

Case 7: Why do customers phone the bank?

A Danish bank wanted to see what its options were for transferring its customer services from the telephone to self-service. This would allow it to reduce the costs incurred in responding to the millions of calls received annually. Firstly, we investigated which customers phoned and why in order to assess if some calls could be moved to self-service.

To ensure that the data was representative, we set data collection for selected days over a month in order to take into account any variation in call types during that period. For example, the end of a month typically sees overdraft requests, while the start of the month sees many calls regarding standing orders/direct debits. It was also important to take measurements at different hours of the day and on different weekdays. For example, many customers called during branch opening hours to schedule a meeting with their advisers.

From the start, we decided that it was important to involve a broad group of employees in data collection (seven from the call centre were chosen). This created project credibility and ensured the quality of the data collected. The employees were relieved of normal duties and began collecting data.

The categories of call types were determined during workshops with management and employees. Afterwards, we tested the categories in a pilot data collection lasting four hours where a lot of possible improvements came to light. For example, we discovered that one of the seven employees did not enter the data from short calls as she was under time pressure. The validity of the data would have been undermined had we only registered the long calls.

Data collection was carried out over the next five weeks and formed the basis of a series of significant decisions on the bank's customer service.

Analyse data

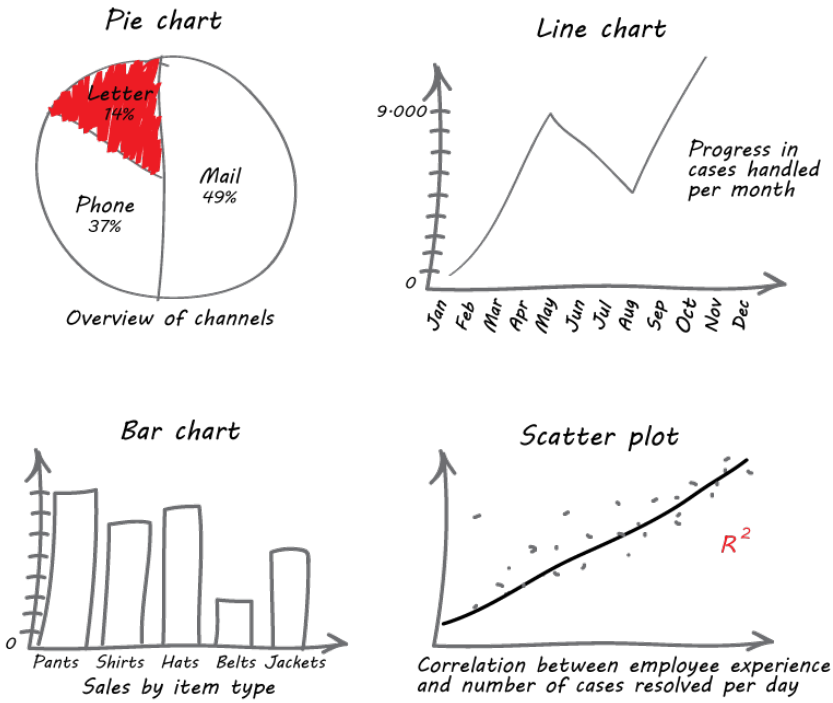
After collecting the data, you must analyse it; focus on which of your X factors influence your Y factors. A wide range of statistical analyses can be applied within Six Sigma — too many to cover here. We will, therefore, concentrate, in this section, on the basic analyses that we use most frequently.

The analysis has basically two steps. Firstly, you identify a statistical connection between the X and Y factors. Then, you explain this connection based on your knowledge of the processes.

Visualisation

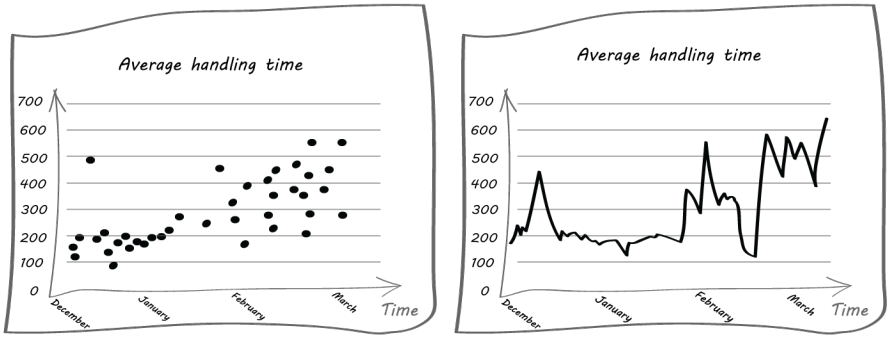
When you have collected the data, plot them to graphically identify any trends. The various types of charts and diagrams which can be used will be covered in the following section. These include:

- Pie chart: shows the distribution of 100% of different categories
- Line graph: shows development over time
- Bar graph: shows comparisons of data categories
- Scatter diagram: shows the relationship between two variables



Examples of graphs used to map data: pie chart, line graph, bar chart and scatter diagram. Diagrams visualise data, and assist in communication and analysis.

Mapping data visually can show patterns and deviations that can help to locate cause–effect relationships. Deviations are interesting because they can provide insight into what affects your Y factor. Even though graphic representations are based on factual data, the type of diagram used to visualise your data can decisively influence how the content is understood. For example, the data from a customer service centre showing the average processing time is represented by two graphs below.



The average processing time at a customer service centre plotted in a scatter diagram and a line graph. The same data can be perceived differently depending on how they are represented.

Even though the graphs show the same data, the line graph makes it easier to identify the monthly variations.

Advanced analyses

Graphic visualisation of data gives a good sense of its coherence and can be sufficient to uncover any issues. In some cases, however, it will be necessary to examine the data more closely. Next we will look at a range of typical analyses which we work with daily:

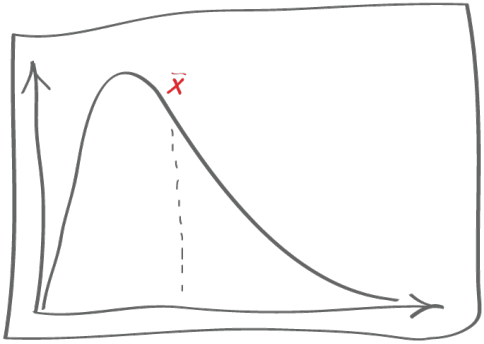
- Data comparison
- Population evaluation
- Development over time
- Degree of influence

Data comparison

Comparison of data is about investigating whether an X factor affects your Y factor. This might, for example, involve asking half your sales team to push a price reduction on selected products to see how great

an effect this has. Afterwards, the average sales for those who promoted the reduction can be compared to those who did not.

When averages are calculated based on a data set, there are several pitfalls to be aware of. An average does not reveal the variation from one extreme to the other. Take for example, customer telephone waiting time; while many customers are on hold for a short time, there is a group who wait for a significantly longer period. If the data shown below is not evenly distributed, it does not make sense to apply the average.



Plot of unevenly distributed data on customer telephone waiting time indicating the average. For such data, the average is a poor reflection of the data.

If we calculate an average waiting time based on the data, it will give an inaccurate picture of customers' experience. In the same way, the average does not accurately represent data distributed over multiple values. Imagine, for example, a customer satisfaction survey based on a scale of 1 to 5 where half of the responses were 1 ('very dissatisfied'), while the other half were 5 ('very satisfied'). Calculating the average of the responses, you will get a customer satisfaction rating of 3, which gives the false impression that customers are neither especially satisfied nor dissatisfied.

Be aware of the amount of data. It rarely makes sense to calculate

the average of a sparse data set. If you have fewer than 20 observations, you will get a better picture by presenting the collated material.

Be aware of how confident you are of your average. If your average is based on random samples, there will be a certain degree of uncertainty associated with it. The three most important factors for uncertainty are:

- The number of observations: the more observations, the greater confidence you can have in the data
- Variation in the data: the less variation there is, the more confident you can be in the average
- A broad distribution of samples in the data

As previously mentioned, the expression of distribution in the data and one of the most used concepts in Six Sigma is standard deviation. Standard deviation is a number that shows how far from the average your observations lie — i.e. the spread of the data set. We use standard deviation to compare different data sets in order to calculate how confident we are of the average. The standard deviation will also show the success criteria for a process being statistically under control. Control refers to how much variation is allowed in a data set and thus how much data is allowed to vary from the average.

Evaluation of groups

Another typical analysis is the evaluation of groups, for example the comparison of the efficiency of employees or the profits of international offices. If you are working with large populations, it is best to divide them into subgroups, as in the case below.

Case 8: Who phones us?

The customer service department of a mobile phone company experienced a sharp increase in calls, leading to increased customer service costs. Management felt that most of these calls were from customers that the company earned very little from and therefore were considering introducing a small customer service charge. Data on all calls from the preceding three months was collected to get an overview of the potential for introducing such a charge. The name of the customer and how much the company earned from them was noted for each call. Customers were then divided into four quartiles (i.e. an equal number of customers in each group) based on company earnings, and the amount of calls was added to the corresponding quartiles. This showed that less than 10% of calls were from the 25% of customers that the company earned the least from. Those that represented the greatest earnings for the company (25%), on the other hand, made 40% of all calls. The company was in fact considering introducing a potentially unpopular measure that would directly affect those customers from whom it made the most profit. The charge was therefore not introduced. Handling calls more efficiently or removing the reasons for such calls, by reviewing their communications with customers in other departments, were instead the focus.

Development over time

Another common analysis method is to investigate when the number of customer service calls or the sales of a certain product increased or decreased. If you examine the data for a certain period and discover where the change began, you can then discover what happened at that

point to cause the change. If sales, for example, have been declining over a certain period, the analysis can help to identify when the decline began. From there, you can investigate whether at that time any changes were made to products, new sales personnel were employed, etc.

Case 9: first call resolution in technical support

The number of first call resolutions (FCRs) —inquiries which could be resolved during the first call — had been decreasing in the Danish media company's technical support department for some time. Data for the previous three months was analysed and showed that the decrease began precisely one month earlier. Following a review of that period, it was proposed that there was a connection between the decrease and the introduction of new staff. The data for the new employees were analysed and two persons were identified as having extremely low FRC rates which were therefore affecting the average for the entire department. Once they had been identified, it was a simple matter of providing them with better training and coaching, and performing ongoing reviews on the quality of their call responses.

Degree of influence

Occasionally you might want data on the degree of influence that a single X factor has on your Y factor. For example, the influence construction site employees' experience has on the number of accidents they are involved in. The data here is grouped in pairs, i.e. for each employee there are two figures: experience in years (X) and the number of accidents within the last year (Y). You then use a correlation analysis to investigate to what degree Y can be explained by X.

Case 10: how long does it take to sell?

Customer service was planning to introduce sales targets for employees. Up until then, employee metrics were primarily quality and call time, but now they would also be measured by how much they could sell in connection with service calls. Employees believed that to increase sales during calls, they would need more time per call. Management, however, preferred to avoid significantly increasing call time.

The department manager collected data on sales percentages and average call time for each employee for two consecutive months. The analysis showed a pretty clear correlation: the shorter the call, the higher the sales percentage. The result surprised many who had expected the opposite and gave rise to a lively debate. Further analysis revealed that all the employees who had short call times coupled with high sales percentages were the most experienced, while those with the least experience had the lowest sales percentages and long call times. The conclusion was that the most important thing for sales was not necessarily extra time but more training for inexperienced employees.

From statistical correlation to actual causality

A statistical correlation does not necessarily indicate a causal connection because the correlation may be coincidental. For example, for many years, there was a statistical correlation between the falling number of births in Denmark and a fall in the number of storks visiting the country. A correlation, which we hope you will agree, is coincidental.

A statistical correlation may indicate the presence of a third factor.

In the example above, for instance, the third factor affecting sales and call time was experience.

Finally, you have to be aware of what influences what. In some cases, it is not your X factor affecting your Y factor but the other way around. For example, it would be easy to create an analysis that showed that the increasing number of psychologists in Denmark over the last 40 years has led to an increase in the number of employees on sick leave due to stress. The causal connection, however, would, in all likelihood, be the reverse.

To determine whether there is an actual cause–effect relationship between X and Y factors, you need to be able to substantiate how the connection is created. For this, you need knowledge of the process.

Understanding and accepting conclusions

It is important to ensure that there is understanding and acceptance of the analysis conclusions before you define any solutions. Six Sigma does not have the same focus on employee involvement as Lean. Therefore, it is important to validate the conclusions with employees who work in the process daily. Another practical way to create acceptance of the conclusions is to involve staff in data analysis. Most employees will have more confidence in the analysis if they or their colleagues have helped to collect the data. If they have not been involved in the analysis, you can involve them in commenting on the results before you collate your final conclusions.

Summary

The purpose of the *measure and analyse* phase was to create a shared analytic picture of the extent and nature of the process problems.

You initially follow the processes involved from A to Z to get an understanding of the process steps and any possible problems.

Then you create a task overview where you map the task in the relevant departments with regard to execution, responsibility for execution and time spent as well as problems in the process.

Next, you create a prioritisation of the problems using a Pareto chart.

After the prioritisation, you chose the appropriate tools to analyse the problems and begin developing solutions. Throughout the *measure and analyse* phase, you remain focused on involving employees and other stakeholders to secure a sense of ownership over the solutions.

Once the most significant causes of the problems have been mapped, the analysis is concluded. This is followed by a process of developing and implementing solutions, which we will cover in the next chapter.

4 · **IMPROVE:** SOLVING THE PROBLEM

In the *measure and analyse* phase, we highlighted the extent of the problem and the underlying causes. Now it is time to develop and implement the solutions. In our experience, the *improve* phase is often the most challenging phase for many projects. In the *define* phase, and *measure* and *analyse* phases, it is natural to focus on listening to stakeholders and giving them the space to vent their frustrations. They can 'dream' of various solutions. The project is still new and open to new input and adjustments. In the *improve* phase, solutions must be made concrete, and we find that many organisations are impatient when they reach this stage. It is also during this phase that the greatest involvement of employees takes place. Old routines are broken and new ones are put in place. The *improve* phase makes great demands on change management, communication and stakeholder involvement. Although people tend to like their own ideas best, there is a greater chance that they will accept novel solutions if they feel they have had a hand in developing them. This feeling of ownership in those who will implement the solutions is essential for success. Communication is crucial because those involved in the change process have a basic need to know what will happen, and when, why and how it will happen; what will be expected of them and what influence they will have. It may seem obvious but we find that these needs are often overlooked. Those who initiate the process have all the knowl-

edge they need and are often subject to time pressures once implementation has begun. Once again, you need to focus on ownership. A significant part of your communication should, therefore, start in the earlier phases. If you have not already managed to create understanding and acceptance of the aims and methods of the project, then implementation will prove very difficult and the chances of success are low.

However, involvement and communication alone are not enough. The most important thing is to make progress and achieve results. That is the aim of the *improve* phase. How to tackle this in practice is described here:

1. Generate ideas.
2. Prioritise the ideas.
3. Sell the recommended solutions to management and the rest of the organisation.
4. Set benchmarks and revise the milestone plan and objective setting.
5. Implement the solutions.

In this chapter, we will review these points and present the tools that we use to structure and facilitate the work.

Ideas for solutions

In our experience, many people have a tendency to start considering solutions once a problem has been presented. It is, of course, often a good idea to address problems quickly. However, if you choose the first solution that you hit upon based on experience, bias or personal preference, you risk overlooking other possibilities which may be cheaper, simpler or more effective. You also risk your personal preferences overshadowing the results of analysis. You can favour a solu-

tion and attempt to fit your analysis to it rather than approaching the analysis with an open mind. It is a waste of time, money and effort to perform a thorough analysis and then implement a solution which has not been shown to be the best.

When deciding upon a solution, firstly you generate a rough list of possible solutions. At this point, which solution would be most advantageous to implement is not important. What is important is to ensure that you think widely so that there are several alternative solutions to choose from. There will almost always be more than the one most obvious option.

If, for example, the problem is too many data entry errors, the options might be:

- To improve employee training
- To improve the IT system's generation of error lists
- To shift the data entry to customers, etc.
-

Our usual rule is that project groups must propose at least three solutions when examining a problem.

Possible solutions can come from many different sources, for example:

- Written into the project description from the outset
- From the best practice of other organisations
- From ideas which emerge during analysis
- From structured brainstorming

No matter where the ideas come from, we recommend a structured brainstorming session to find further ideas or develop existing ones.

Brainstorming generates a wealth of ideas

The purpose of brainstorming is to generate a large number of ideas based on synergy, i.e. that participants inspire each other, build on each others' ideas and together suggest more and better proposals than what they could come up with individually.

When conducting a brainstorming session, there are a series of ground rules that it is important to agree on initially:

- Stick to the topic.
- Give space to others to discuss and question.
- Build on each other's ideas, no one 'owns' an idea.
- Be concise.
- Avoid taking a stance on ideas during the session, no matter how unsuitable you might think they are.
- Make room for silly or frivolous ideas. They can often be the inspiration for creative and usable ideas that would otherwise not have come to light.
- Be positive about other people's ideas.

It is also important to consider the framework of your brainstorming session. You must ensure that your group consists of the right people with the relevant professional and personal skills as well as the motivation to be creative. It is also a good idea to think in terms of employee ownership — those who will later implement the solutions. It can be advantageous to bring in an external facilitator who does not have an interest in the topics under discussion. This also gives you, as a leader, the option of participating in the discussion or taking a more withdrawn and analytic role. Finally, you should set a time frame for how long you have to generate ideas. It is sometimes a good idea to create time pressure to get participants to let go of the limitations they might otherwise place themselves under.

You can choose from several methods to encourage creativity during your brainstorming session. It does not matter which one you choose as long as you choose one that suits you and the organisation. A review of three methods that work well for us is given below.

Negative brainstorming

Start by formulating the problem, for example: 'Our case handling often has errors'.

Next, begin to generate ideas for how the problem could be made even worse. For example, by answering the question: 'How could we make sure that our case handling has even more errors?' This will result in a long list that might look like that shown in the figure below.

Create a culture which disregards customer wishes.

Evaluate employees on productivity alone, not quality.

Fail to train employees

Ensure there is no opportunity or space for knowledge-sharing among employees.

Publicly humiliate anyone who admits to making a mistake.

Fail to investigate why errors occur.

List resulting from the question 'How could we make sure that our case handling has even more errors?' posed during negative brainstorming. Negative brainstorming gives many ideas and and also focuses on areas that are currently performing well.

When you have generated a list of proposals for making the problem worse, flip those proposals around and assess them as possible solutions.

Proposed negative idea	Possible solution
Create a culture in which we are indifferent to our customers	Create better understanding among case handlers for the ways in which errors affect customers Closer customer contact
Measure employees purely based on productivity rather than quality	Introduce quality targets for case handling Revise efficiency targets
Fail to train employees	Evaluate introductory training Implement better ongoing training and refresher courses
Ensure there is no opportunity for knowledge-sharing among employees	Encourage knowledge-sharing Facilitate knowledge-sharing — provide the physical space, time and rewards
Publicly shame anyone who admits making a mistake	Ensure that management create a culture in which it is okay to own up to mistakes
Fail to investigate why the mistakes happen	Implement a process for ongoing error prevention

Examples of ideas from negative brainstorming that are reversed to become possible solutions.

There are several advantages to negative brainstorming. This is partly because it is easier for most of us to think negatively than positively, when we need to we can come up with more suggestions for making things worse than for making them better. In addition, negative brainstorming focuses us on the things that work well and do not need to be changed. We might never have discussed these if our brainstorming had been positive from the outset. Lastly, it is a fun exercise, encouraging creativity, creating good energy and resulting in some amusing suggestions.

Challenging assumptions

Start by formulating the problem, which might be: 'We use too many resources compiling monthly reports. They often arrive late, and are full of mistakes'.

Write down as many of the existing assumptions of the problem as possible in the first column. Then try to challenge each of these assumptions in turn.

Assumption	Challenge	Proposed solution
We have to compile a monthly report	Do we have to compile a monthly report?	Provide continuous access to key performance indicators (KPIs) instead
It has to contain 48 KPIs and analyses	Are all those KPIs and analyses really necessary? Which ones are used?	Eliminate unused figures and analyses Make it possible for management to extract the figures they need
It has to be delivered within the first three working days of the month	Is it important to deliver it on that specific date? When are the figures actually used?	The report is split so that the most important KPIs are reported first and the rest are sent on later
It has to be printed and distributed to every person in senior management	Is it important to print it?	The digital version of the report is available on in tablet readable format Continuous access is provided to KPIs

Examples of assumptions, their challenges and the solutions proposed. Challenging assumptions can provide a new perspective on familiar problems.

The advantage of this method is that you can question an organisation's sacred cows which most people never question in their day-to-day work. Nevertheless, employees often have difficulty in objectively addressing the assumptions in their processes because the process

stages have become routine and therefore ingrained. In these instances, an external facilitator can be used to ask critical questions and facilitate an alternative conceptual approach. In addition, brainstorming can be supplemented by your work on value stream analysis and your list of value-adding activities. These tools are useful for revealing unproductive assumptions.

White spaces

Solutions can have either short- or long-term value. We mentioned quick wins earlier and they are especially appropriate here where creative and objective approaches to problems can result in some immediately beneficial solutions. However, long-term solutions are also important. By separating solution proposals into short- and long-term perspectives, you can begin prioritising proposals by potential effect. Note all the proposed solutions to the problem(s) which are hampering the process (see below).

Here is an example of a problem: 'We spend too much time handling cases because the printer is too far away and it takes five minutes to fetch printouts'.

Add possible solutions to the table under the short- and long-term columns, as appropriate. These solutions should cover the physical equipment and materials of the company, human resources, external factors and environmental considerations.

Solution category	Short term	Long term
Equipment (physical equipment: IT, shelving, layout, etc.)	Purchase an extra printer	
Process (the way work is carried out)	Move the printer closer to the employees Fetch printouts twice a day	
People (employees, customers, partners, etc.)		
Materials (physical materials or information that is used)		
Leadership (how is the task managed)		
Environment (external factors: legislation, media and interest groups)		
Measurement (how to measure process performance?)		

Brainstorming table (first phase) based on the problem: 'We spend too much time handling cases because the printer is too far away and it takes five minutes to fetch printouts'. Note that blank spaces are included. Brainstorming with blank spaces can ensure a breadth of ideas.

Next look at the solution categories where blank spaces are present indicating no suggested solution. Focus on filling in these spaces by challenging proposed solutions and brainstorming once more.

Solution category	Short term	Long term
Equipment (physical equipment: IT, shelving, layout, etc.)	Purchase an extra printer	Deploy email support so that we can email customers instead of sending a letter
Process (the way work is carried out)	Move the printer closer to the employees Fetch printouts twice a day	Phone customers instead of writing
People (everyone involved in the process: employees, customers, partners, etc.)	Ask all employees, when collecting their printouts from the printer, to deliver finished printouts to their colleagues	Outsource printing and packaging to an external supplier
Materials (physical materials or information that is used)	Ask the customer to indicate if they would like a response to be sent	Establish electronic communication with customers via the company website
Leadership (how is the task managed)	Focus on better communication with customers so that fewer inquiries arise that require responses	Emphasise that email responses are part of the company's environmentally friendly profile
Environment (external factors: legislation, media and interest groups)		
Measurement (how to measure process performance?)		Measure the quality of the answer rather than the time used; perhaps savings can be made by writing better responses

Brainstorming table for the second phase of brainstorming based on the problem: 'We spend too much time handling cases because the printer is too far away and it takes five minutes to fetch printouts'.

It is not always possible to find solutions to fill in all the blanks. However, new ideas will often come up when reviewing the table. This therefore ensures a broad perspective and a focus on both long- and short-term solutions.

Prioritising possible solutions

After generating proposed solutions, the next step is to sort them. There are a number of different ways to prioritise solutions. From a prioritisation perspective, it does not necessarily make a great difference which method you choose. From a change management perspective, however, it can be crucial to choose the method most acceptable to the organisation. Paradoxically, greater resistance to the change is often created if the selection process is perceived as arbitrary or subjective, rather than implementing a solution with an unattractive result but where the process is viewed as having been unbiased.

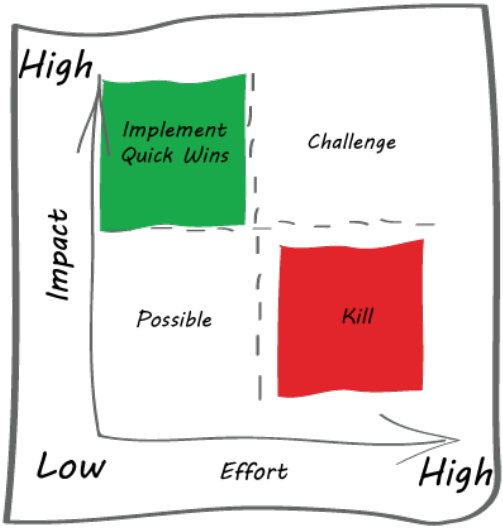
We often succeeded in prioritising proposed solutions based on the following methods:

- PICK charts
- Voting n/3
- Prioritisation of Y factors

Next follows a short description of these three methods and their advantages.

PICK chart

The PICK chart is a simple ranking of the solution options. It is used to assess the relative effort required to implement each solution and the relative impact of each solution. Proposed solutions are separated into four categories: possible, implement, challenge and kill.



Possible: might be worth implementing

'Possible' characterises solutions that are easy to implement but have no major effect. These may be worth implementing, however, if they are visible for employees, customers or management. These types of improvement are often numerous and their combined effect over time can be considerable.

Implement: quick win

This category is for solutions that are easy to implement and have a major effect, i.e. quick wins.

Challenge: might be worth implementing

This category covers solutions with a large impact but which require a large effort to implement. These could be the introduction of new IT systems, wide-ranging organisational changes or changes in the culture of the organisation. It is important to challenge these solutions and assess whether their impact is worth the effort.

Kill: not worth pursuing

As the name indicates, it is best to disregard solutions that fall into this category. These are the solutions that are difficult to implement and have little effect. The effort required to implement them does not result in any significant returns. So these solutions should be disregarded or reformulated.

The value of the PICK chart increases as the number of stakeholders involved in completing the chart does.

The most important thing when using the PICK chart is the degree of completeness. The stakeholders who will own the future process must feel that they are a part of ranking the possible solutions with the project team who have created the detailed analysis.

The evaluation of the effort and impact of a given solution will, of course, be subjective. Therefore, it is a good idea to ask participants to assess possible solutions individually and then compare their assessments. If these assessments vary greatly, the solution should be reviewed again. The variation may reflect varying ideas of the solution's content and needs to be explained in depth.

It is a good idea to compare the grade you have awarded each idea to see if some should be adjusted.

Voting n/3

Another way to rank proposals is to gather the stakeholder group and ask it to vote on the possible solutions. This is typically done by giving stakeholders a certain number of votes equal to the number of solutions divided by 3. The idea of giving participants this number of votes is to push stakeholders towards prioritising what is most important. After the first voting round, you discard those proposed solutions which received no votes.

If more proposals remain than it is feasible to implement, it is a good idea to have another round of voting. You can also award nega-

tive votes to proposals which participants do not want. It is not crucial to divide by three. The most important thing is that participants have sufficient votes to ensure that they do not simply vote for their own suggestions. On the other hand, they should not be given so many votes that they have difficulty casting them all.

The advantage of this method is that it creates a picture of which proposals have the broadest support in the organisation. In our experience, the probability of success is greater for solutions that have broad support. We would rather implement a solution that is 75% perfect but has 100% support than implement one that is 100% perfect but only has 75% support.

You should check that each proposed solution is practically possible before beginning the voting process. Before voting, then, the project group should discard those solutions which:

- Are deemed too costly
- Have been rejected by management

Contravene legislation, regulations, etc.

Prioritisation: Y factors

Prioritising with the help of Y factors often works well if there is disagreement in the project group on how to weight the proposed solutions. This is done by listing the most important Y factors that the solution must take into account (see section 3.4 on Y factors). The table can help in the ranking process by highlighting the importance of Y factors enabling participants to avoid assessing the proposed solutions based on instinct. In addition, completing the weighting and awarding points facilitates a good discussion of what is important to the organisation and the comparable strengths of the various proposals.

We find that this method works well when the organisation needs to ensure a visible prioritisation of the solutions.

Y factors typically fall into the following categories:

- Customer satisfaction
- Profitability
- Employee satisfaction
- Other (e.g. company image, public relations and compliance with legislation)

Give each of your Y factors a weighting from 0 to 10 depending on how important the various factors are for the project.

After this, review the solutions one by one and weight them from 1 to 10 in relation to each Y factor. Calculate the total score for the individual solution by multiplying its points by the weighting given to the individual Y factor and total the result. For example, for solution B below, the total score is (2 × 10) + (0 × 7) + (10× 5) (0 × 0) = 70 (see below).

	Customer satisfaction	Profit	Employees	Other	Score
Weighting	10	7	5	0	
Solution A	8	5	0	0	115
Solution B	2	0	10	0	70
Solution C	0	8	3	0	71

Example of the weighting and ranking of Y factors. This helps you to select proposed solutions.

Sell the solution to the steering committee

When you have analysed the problem with the project group, found the underlying causes, assessed the possible solutions, ranked these solutions and identified those you will recommend, the last step before implementation is to sell the solution to the steering committee.

Even the best solutions are worthless if you cannot persuade the decision makers and other stakeholders of their value. Months of analysis can be wasted if you do not communicate the results effectively.

The steering committee and stakeholders often do not know the detail of the analysis and solutions. They have not got the same understanding of the project as the employees who took part in the *measure and analyse* phase. You will often only have a short time to present the proposed solutions and the considerations that underpin them to decision makers. Therefore, it is important that you communicate effectively and base your selling strategy on the decision makers' interests, concerns and situations.

We base our communication with management on the principles discussed below.

Know your audience

Put yourself in the position of each steering committee member, their focus, problems and concerns. They will often be more interested in results than methods. Draw up an overview of how much you expect to earn or save using the solution and when you expect this to happen. Include a rundown of the precise costs of implementing the process improvement.

Express your view of the impact on the organisation, including the political implications.

Will this alter the balance of power between departments and management? Are there some parts of the solution that managers will find hard to sell to their parts of the organisation?

Communicate the overall view and concrete solutions

Present the solution's key figures (savings, increased sales, etc.) and the conclusions as an outline. However, keep your notes on estimates and calculations handy so you can answer any questions that arise.

Give an overview of the major elements based on:

- The project plan, including the most important implementation tasks and deadlines
- Stakeholder analysis of the most important parties involved and their responsibilities
- The solution's business case of costs and benefits

Visualisation of analyses and solutions

Make the analyses and solutions easier to understand by visualising them. You could do this by using graphs, which show correlations, tendencies and variations, or diagrams such as flow charts, swim lane diagrams or spaghetti diagrams. For example, the solution can be visualised using a diagram or storyboard that relates how customers and employees experience the new solution.

No surprises

One risk factor in the presentation is your audience's interpretation of what you present. Formulations might be misunderstood, messages can generate resistance, questions may be asked about details and conclusions, and suddenly the presentation and the decision-making begins to move in an unexpected direction. In the worst–case scenario, the solution will be rejected or you will be asked to work on it some more. This is why it is important to ensure that there are no surprises for key decision makers in the presentation. Review critical points, conclusions, recommendations and formulations with key decision makers before the meeting so you can avoid potential conflicts. Ideally, they should sit and nod in agreement when you present your analyses and conclusions.

Anticipate questions

Try to anticipate the concerns of the various stakeholders. What questions are they likely to pose given their interests and responsibilities? If you can anticipate questions, you can prepare answers and other back-

ground material to bring with you. If you are confronted with questions that you cannot answer, the decision could be postponed or you could seem uncertain of the solution and fail to convince your audience.

Present concrete decision options

The challenge of the presentation is to persuade decision makers to accept the proposed solutions. So be specific and precise about what you want them to accept. You can break the proposal up into phases so that the steering committee can give an initial green light at the presentation and afterwards decide on the remaining elements.

Be prepared

The first time you present a message, you will almost always be more uncertain than on the second, third or fourth occasions. Rehearse the presentation in front of an audience — preferably those familiar with the project details as well as those who know nothing about it. This will hopefully give you usable feedback on your explanations, timing and style of communication.

Prepare an executive summary before the meeting so that participants can show it to colleagues and stakeholders they work with that might need persuading.

From decision to implementation

When your project group has received the green light from the steering committee, the next step is to prepare for implementation. Firstly, draw up a list of check points at which the implementation should be analysed. Next, compose a list of tangible goals — what results should be reached and when. Included here is the process for documenting those targets being met. These check points are important because they function as milestones marking change. You can use them to navigate implementation and adjust your efforts. In addition,

they indicate a clear path for communicating the project. Finally, they can be the basis for showing the impact of the project, celebrating its success and creating support and enthusiasm for the project.

The check points are based on the target outcome that you set in the define phase. At this stage of the project, you know precisely which solutions you are pursuing, which have realistically achievable effects, and where and when it will be pertinent to follow up on the effect. Therefore, work through the target-setting thoroughly and set precise check points for the implementation.

We work with various target levels because it is not possible to measure everything immediately after implementation. We first measure if the initiative has been carried out. Then we measure the reaction to the initiative, etc. The table below gives an outline of these target levels.

Check point levels	**Focus**	**Suggested measurement methods**
Initiative	Have we carried out what we intended to do?	Simple documentation of the executed initiative
Reaction	How has our initiative been received? Has it been supported?	Satisfaction and opinion surveys in the immediate aftermath of initiative
Learning points	Have employees mastered the necessary skills to carry out the new processes?	Test of partial skills and test runs of the processes
Behaviour	Have the new processes been adhered to?	Observations of the process steps
Effect on the organisation	Does the new process have the desired effect on the organisation?	Summary performance goals
Return on investment	Was the impact as valuable for the organisation as expected?	Summary of financial goals

Examples of target levels used to evaluate solution implementation. The implementation is evaluated via check points on several levels.

Remember that all your check points must fulfil the SMART criteria. When they have been set, work them into your milestone plan. Then work through the milestone plan as described in the *define* phase. If surprises occur — e.g. solutions that are more expensive than budgeted for, scheduling conflicts, targets that turn out to be unrealistic, etc. — then it is important that you involve the steering committee and get a clear mandate from it to adjust solutions, schedules, targets, etc. In the next section, we will describe how to implement solutions in practice.

Implementing changes

The most important principle to remember when implementing process improvements is that it is a continuous process (see section 2.8). We rarely want to wait to analyse our way towards the perfect solution but rather choose to implement the solution that is 80% right, evaluate it and then adjust and improve it and work towards our goal. Implementation as an ongoing process has several advantages, including the following:

- Quick wins: we implement improvements immediately.
- Trial and error: we can test different options and find the one that works best.
- Dealing with uncertainty: instead of analysing all consequences and parameters, we implement a solution and follow it closely.
- Shorter implementation times reduce the risk of circumstances, and indeed the problem itself, changing, while we spend time analysing it.

Part of the continuous approach is to begin with a pilot programme, i.e. a limited implementation that can be withdrawn if it doesn't work. This might be basing the implementation in a single location, or for

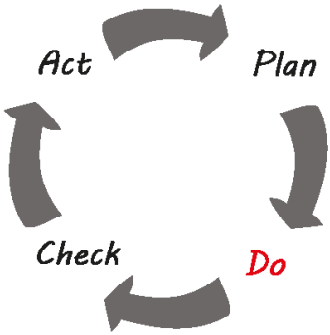
a specific set of employees or customers, etc. Beyond creating an ongoing process around the implementation, another important principle is to avoid too many concurrent changes. We generally prefer to implement our improvements and check that they have been realised and function properly before we begin on the next implementation. This has several advantages. Most organisations have difficulty in handling too many changes at once. If we try to implement too many concurrent changes, we risk encountering resistance, loss of overview and change fatigue in general.

Many implementations will draw on the same resources in an organisation. Implementing too many changes at the same time risks a lengthy implementation time and delayed implementation returns.

We facilitate the implementation of changes using the Lean tool Plan–Do–Check–Act (PDCA). PDCA consists of four phases of implementation:

- **Plan:** plan the implementation before, during and after.
- **Do:** implement the idea in a limited area of the organisation.
- **Check:** evaluate the implementation and the impact of the idea.
- **Act:** adjust whatever is necessary. If something is not working, it needs to be changed. Weigh up whether this will lead to further returns.

The PDCA cycle is often depicted as a wheel (see the figure below).



Example of a Plan–Do–Check–Act (PDCA) wheel. The PDCA wheel is used to ensure that implementation goes through all important phases.

The implementation model — the PDCA wheel — is also called the quality wheel. In Toyota, W. Edwards Deming used it for minor changes in production. The idea was to have a quick process for evaluating small changes without carrying out large and costly analyses. Change implementation is a large and important part of a project. However, we will not review it in detail here as implementation involves general management, project management and other disciplines not necessarily unique to process improvement. Added to which are the significant differences in approach, depending on whether you need to implement a new IT system, introduce new production machinery or make changes to the organisation's structure, etc.

Summary: conclusion and learning points

We introduced this chapter with the observation that the *improve* phase makes significant demands on your ability to manage change. In our approach, involvement and communication are an important part of the solution. As a rule, employees and managers prefer their own solutions, therefore, you should involve them in the design and

selection process. Also remember, the employees and managers who have not been involved need to know what is going to happen, and why, when and how it is going to happen as well as what is expected of them.

However, communication and involvement alone are not enough. The most important thing, of course, is to ensure progress and results, which was covered in the rest of the chapter.

The first step in this aim is to generate several possible solution ideas so you can pick and choose between them. We presented different brainstorming methods to encourage idea generation. The next step was to select ideas. Once again, it is important to choose a method that suits you and is supported in the organisation. We presented the PICK chart, voting n/3 and ranking Y factors.

When you have described and chosen your solutions, the steering committee must approve them. In relation to this, we devised seven principles on knowing your audience so as to increase the probability of gaining the steering committee's approval to pursue the project.

When the steering committee has given your project the green light, you return to planning. The first task is to set check points for the implementation and changes. You should, in this order, measure:

1. If the agreed initiative(s) has/have been carried out
2. What the reaction to that initiative(s) has/have been
3. If the relevant skills have been developed
4. If the new process steps are being followed
5. If the organisation has experienced the desired effect
6. If possible, what the likely return on investment will be

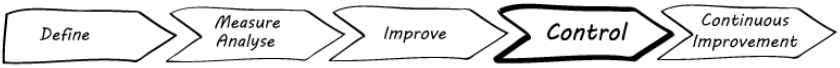
When the check points have been set, go back to the milestone plan and revise your goal setting. If unexpected changes occur, discuss the issues with your steering committee.

The implementation itself typically follows the PDCA wheel. Often

we do a pilot implementation to test the solutions before involving the entire organisation.

With the implementation, the *improve* phase is concluded. Now all that is left is to document the effect of the implementation and to ensure that the changes are maintained. We will discuss this in the next chapter.

5 · CONTROL: FOLLOW-UP ON PROBLEM-SOLVING



Control is an extension of the implementation of the changes. The purpose of the *control* phase is to complete the project; hand it over to those operationally responsible; and develop a plan and standardisation so the organisation benefits fully from the returns of the project. It is of central importance to establish routines and follow up on them in order to make sure that employees do not revert to old patterns of process handling.

In the previous chapter on the *improve* phase, we argued that implementation should be an ongoing process. This also means that the transition from the *improve* phase to the *control* phase is continuous. There are often many balls in the air at the same time and responsibility shifts from running the project to running the operation, which requires time for handover. The validation of the results and financial gains of the project will be carried out several months after the official project conclusion. It is often only at this point that it is possible to assess the actual results. Therefore, it is important to maintain an overview of the project. This is where you reap the benefits of the detailed project management efforts you made earlier, especially your milestone plan with check points.

The *control* phase must ensure that:

- The solutions that are implemented have been correctly put in place, and that employees maintain these practices and new processes after the project group have left.
- The solutions have the impact you wanted. Perhaps there have been unforeseen side effects or there are conditions you did not take into account in your planning.
- You get the maximum return from the project. Can the solutions be developed further? Can they be applied in other areas of the organisation?
- The results and experiences are documented and communicated to others in the organisation.

All of these issues will be looked at in this chapter.

Measuring and maintaining behavioural changes

In the previous chapter, we described how we set up check points for initiative, reaction, learning and behaviour. We also described how there was a time lag between the various check points. In the *control* phase, you will typically follow up on initiative, reaction and learning. You will also follow up on whether the new processes are being used.

Many people find it difficult to change routines. Some will feel that their performance worsened for a while, until they executed the new processes so many times that they began to feel natural. The process should be examined in all situations — both every day and extraordinary. Finally, there will always be situations which the project did not take into account in advance. That is why it is important after implementation to follow up on how employees and managers carry out the processes.

There are many ways to check if the processes have been implemented properly:

- Observe the work processes: the Gemba principle which we talked about earlier in the chapter on *measure and analyse*, works in all phases, and one of the best ways to find out how employees will tackle a task is to observe how the concrete task is tackled.
- Interview employees and managers: how have the solutions been implemented, how do the new processes work and have they been developed further?
- Check the documentation of processes: if the changes have not been incorporated into the documentation, they will most likely be forgotten in the long term.
- Check if employees understand why the processes were changed and what the advantages of the new processes are. One of the best questions you can ask is 'What are the advantages of the new process?' If employees cannot list and explain the advantages easily, there is a significant risk that they will not continue following the new process.

Your baseline assumption should be that adjustments will be needed, hopefully small ones, so, at this stage, you can go one more time around the PDCA wheel. However, in some cases, large changes will be needed that will, in practice, cause you to rerun the *improve* phase for that solution.

Follow up on organisational impact

To check the effect of the solution on the organisation, you can use the target outcomes that you formulated as check points during the *improve* phase as your baseline.

What have customers, the organisation and employees gained from the project?

If the check points are SMART-formulated, it will be easy to eval-

uate when they have been reached. Watch out for any unforeseen results of the project — positive or negative.

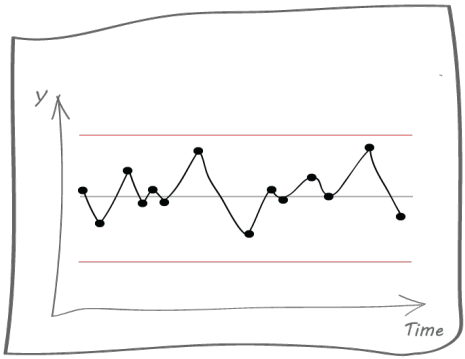
It is often a good idea to follow up regularly on how the new processes are performing. A commonly utilised tool in Six Sigma for this is the 'control chart'.

Use control charts to evaluate the process's performance

The idea of a control chart is to continuously observe the development of processes that require improvement. Control charts can show if processes are performing as intended, if they are successful in their effects and if there is potential for further improvement. They can also help you to keep an eye on progress in general.

A control chart is first and foremost a statistical tool to help you to distinguish between incidental variations in the process and lasting changes. Control charts can be made in many different ways. Common to all control charts is that the overarching parameter for improvement (Y) is shown on the vertical axis and time is shown on the horizontal (X) axis (see the example below). The unit of time can be days, etc., depending on what makes sense in the process. Data for use in the control charts is typically extracted from IT systems. The Y parameter can be lead time for cases or products, average costs per production unit or quality targets.

Then you set the limits for how large a variation your customers can tolerate in your Y parameter. These are represented by red lines in the graph below. The tolerance can be set with the help of the customer's voice as described in chapter 2 above. The customer's voice can indicate how great a margin there may be on the delivery of goods, invoicing errors, etc. Remember that customers can also be internal. For example, they can be managers who are responsible for the economic performance of the processes or employees, departments, etc., that use the output of the process as input for their own processes.



Example of a control chart with red lines indicating the limits of customer tolerance. It is used to continuously keep an eye on process performance.

Now you have a tool to continuously follow up on how the established processes perform, if the project reaches its target outcomes, and whether the impact of the changes is sustained. Once again there may be a need to adjust the solution either with the help of the PDCA wheel or by repeating steps from the *improve* phase. In rare instances, there may be a need to roll back changes and return to the old processes.

Maximum return

Another aim of the *control* phase is to secure maximum return from the project. This can be done in two ways:

- Reuse the process improvements in other areas of the organisation or for other tasks.
- Optimise the developed processes.

Reusing process improvements

If the organisation can reuse process improvements, for example, 'print to pdf file and store electronically', it is possible with relatively little effort to create a large impact.

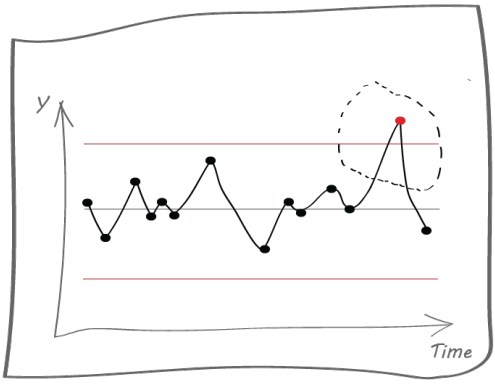
The solution has a value in itself, but you should not underestimate the experiences gained from its implementation, as well as the fact that you can point to examples of the solution working well. This will be of help both in any future decision-making process and in promoting the solution in a new implementation.

To identify opportunities to reuse the solution, research whether any of the stakeholders from the previous stakeholder analysis would find the solution relevant to their needs. In connection with the implementation, other people in the organisation might point out similar issues in other processes. There may be several areas in the organisation where it might be worthwhile to spread the solution.

Optimise the developed processes

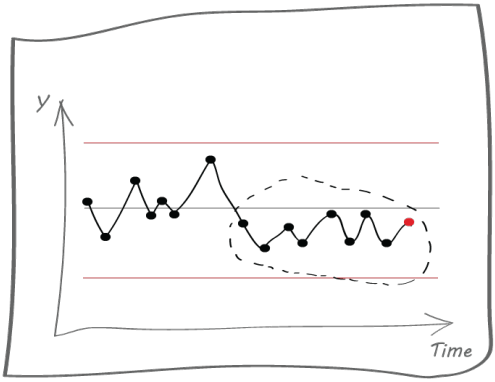
In addition to using control charts to document the effect, you can use them to continuously optimise your processes. The Y-parameter will be the problematic issue that you will want to optimise. Then you should calculate the average of at least 20 consecutive observations and enter the results as a horizontal line. This line functions as a baseline for optimisation.

After that, insert an upper and a lower limit for the probable variation from the average. The value of the upper and lower control limit is, as a general rule, calculated by adding 3σ to the mean and subtracting 3σ from the mean, respectively. 3σ on either side of the mean value gives 6σ. You should, however, know your data material before you apply this method.



Example of a control chart showing when the process goes beyond what is acceptable.

If you observe measurements outside the control limits, this is likely to signal that there have been changes to the process or other extraordinary occurrences which have affected the process. Therefore, it is interesting to investigate the cause. Check the fishbone diagram mentioned in chapter 3 above to get some idea of which X factors you should look for changes in. It is interesting to investigate all forms of variation, both those the organisation sees as positive and those perceived as negative.



Example of a control chart showing when processes are not performing as expected.

There are other signals that are worth investigating, for example:

- A sequence of observations on the same side of the mean (nine in the example above)
- A lasting reduction or increase in variations in relation to the mean

Common to all these signals is that they are statistically improbable and therefore indicate that there has been a change in the process or that some factor is influencing the process. For each signal, there is the potential for improvement.

Final report and return on investment

To conclude the project, you will prepare a final report. It should contain:

- Documentation of handover agreements with employees and management, who now own the operational processes
- Documentation of the project's impact — both intended and unintended
- A summary of the project's most important lessons

At the conclusion of the project, it is important to compare its outcome with the targets in your business case. What was the total return on investment? Did you achieve the outcomes you aimed for? Was it within budget and on time? Was the economic justification correct? We have found that it is important not to assess the return on investment on purely financial parameters. In the section on target-setting in the *define* chapter, we argued for the importance of balancing the target between economic, customer-oriented and employee-oriented criteria. The same is true of return on investment. The project must be

assessed in its entirety or else you will see only a fraction of its effect.

In your summation of lessons learned, it is a good idea to focus on advice relating to the process in your next project. In the course of every project, you will *always* formulate good ideas for new relevant projects and it is worthwhile communicating these. Finally, you will have almost certainly identified possible initiatives for ongoing process improvements. This will be discussed in the next chapter.

Summary: conclusion and learning points

The purpose of the *control* phase is to ensure that solutions are correctly implemented and that changes are maintained. In addition, you need to ensure that the organisation received the intended return from the project. However, the possibilities of amplifying the effect should be exploited by reusing the solutions and optimising the established processes further. The final step in the phase is to conclude the project formally with a final report and a review of the return on investment.

In this chapter, we argued the importance of employees and managers following the new processes after implementation. We described how you could use the check points from your milestone plan to schedule your follow up. (Typical efforts here include observation of the processes and interviews with employees and managers.)

We also described how you could follow up on the effect of the new process for the organisation partly by using the target outcomes that, based on SMART criteria, you formulated during the *define* phase and refined in the *improve* phase, and partly by using control charts.

Both your follow up on changed behaviour and the effect for the organisation would probably reveal the need for minor adjustment to your solutions. You could then implement these adjustments with the help of the PDCA wheel. In some cases, the required adjustments are so great that you will need to rerun steps from the improve phase. The

worst case scenario is that the solution would have to be withdrawn.

We argued that you could increase the impact of the project by transferring the solutions to other tasks or departments in the organisation. In addition, we described how to use control charts to optimise the established processes.

Finally, we detailed how to conclude the *control* phase by documenting agreements with process owners and the impact of the new process, summarising the business case, and communicating the projects most important outcomes.

6 · CONTINUOUS IMPROVEMENT

As mentioned in the book's introduction, work on improving processes can be done as a project or as part of everyday operations. Organisations will often deploy one or more projects to improve processes and solve specific problems, but sooner or later it will make sense for ongoing improvement to be made a part of operations. In other words, every employee focuses on improving processes and solving problems in their daily work.

Establishing ongoing improvement is about realising that perfect processes do not exist and that processes must be fine-tuned every day.

In Lean, this sort of continuous improvement is called *kaizen*, the Japanese for improvement. This is precisely the aim of Lean — to establish a culture of improvement within an organisation. Everyone within an organisation is responsible not only for their own work but also to constantly consider how to simplify or improve things or reduce costs.

Organising operations and improvement

In many organisations work is split into an operational organisation and a project organisation. The project organisation takes care of the strategically forward-looking activities. By separating operation

and improvement efforts, the individual employee is granted a clarity of focus that gives routine to the tasks they are responsible for. The immediate disadvantage of this organisational split is that employees with a brief to improve processes do not possess detailed knowledge of the processes and therefore need to spend a lot of time getting to know them.

The current tendency is for organisations to maintain a separation between operation and improvement, but to ensure cooperation and understanding by exchanging employees between groups at appropriate intervals. The organisation thus makes sure that employees continually switch between performing tasks and reflecting on why their work is carried out in the way that it is.

Another tendency is for departments to be organised in accordance with the core processes. An example of this would be sales, deliveries and staffing being amalgamated into one department that deals with customer enquiries from start to finish. This would break down a range of silos that can arise when cases cross the barrier between different functions.

Lean whiteboards provide an overview and the possibility to react quickly

Lean whiteboard meetings are often the most visible signs that an organisation is working with the process. The whiteboards are central to the implementation of Lean because management can keep track of how many departments hold lean whiteboard meetings. A whiteboard meeting is also the place where employees are regularly confronted with Lean. The content of the whiteboard is designed based on what the organisation needs to focus on. In this way the whiteboards can contribute to improving the organisation's meeting culture and behaviour.

Whiteboard meetings are a quick and an effective way of checking

on the status of ongoing tasks and involving all employees. This provides greater support for the solutions.

The three main reasons for working with whiteboards are discussed below.

Whiteboards boost knowledge-sharing

Whiteboards can boost knowledge-sharing because they create an overview of and a focus on the tasks, milestones or ideas being worked on currently. The overview is partly for the sake of management but it can also prompt exchanges between employees regarding concrete problems. By working with whiteboards at different levels in the organisation, upper management can gain an overview of the most critical tasks, milestones or ideas that are being worked on in the organisation.

Whiteboards create structure

Whiteboards can help to establish a fixed structure, agenda and frequency for meetings during which the most important tasks and issues that might arise in connection with these can be reviewed. The structure also helps to create focus because anything that is not on the whiteboard is not important enough to spend every employee's time on and should be reviewed later.

Whiteboards assist the manager

As a management tool, the whiteboards can function as an agenda for inexperienced managers. By reviewing the points listed on the whiteboard, the department will discuss the most important topics and problems, and, if the whiteboard has been created properly, no significant areas will have been left out. For more experienced managers, the whiteboard will be a tool to ensure a fair process during which meeting participants are included in the decision-making.

Whiteboards should always be backed up by a clear process which

determines when and how information is updated, and identifies who is responsible for facilitating the meeting and what to do if the person responsible for the whiteboard is absent.

Three types of whiteboards are used:

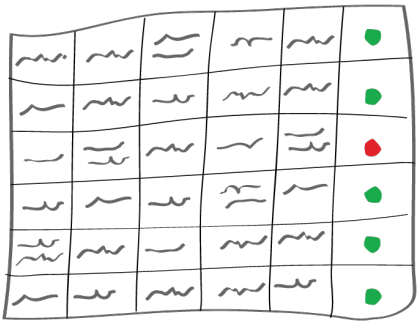
- Operational whiteboard
- Improvement whiteboard
- Target management whiteboard

The appropriate content and utilisation of each type of whiteboard will be covered in the next section.

Operational whiteboard

The operational whiteboard is based on daily departmental tasks. It can be made by carrying out an analysis of existing tasks, preferably in the form of a task outline.

The purpose of the operational whiteboard is to provide an overview of each employee's daily tasks and to continuously ensure that adequate resources are allocated to each task.

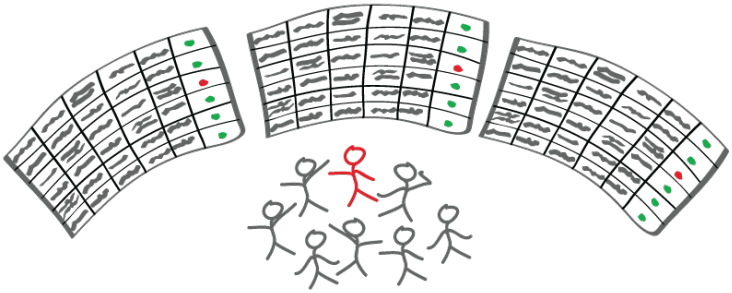


Example of an operational whiteboard. Tasks and their details are listed in the columns, with the dots in the last column indicating the tasks' status. An operational whiteboard provides an overview of ongoing tasks for everyone.

The columns of the whiteboard contain answers to the most important questions:

- What is the purpose of the task?
- How must the task be prioritised during busy periods?
- How is the task shared among employees?
- Who can perform the task?
- How often is it carried out and how long does a single task take?
- What is the status of the task (in the form of red or green marks)?

To ensure that data from individual departments is collated, aggregated and communicated cross-departmentally, weekly meetings are held in a common whiteboard room where each departmental head and project manager delivers a brief and precise status update on performance.

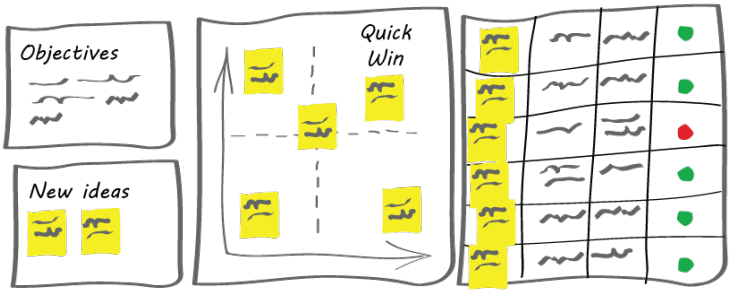


A whiteboard room with the organisational whiteboards of departmental heads and project managers provides the entire organisation with a shared overview.

A manager will often arrange this weekly meeting, which ensures that managers and employees across departments and projects are updated and know what is important for each individual department.

Improvement whiteboard

The purpose of the improvement whiteboard is to take note of ideas that employees might have and to ensure they are implemented. The ongoing focus should be on carrying out tasks in the smartest and most appropriate way. In addition, the improvement whiteboard helps to establish a fixed process for working on improvement proposals. Most improvement whiteboards will contain an element of the PDCA wheel as described in chapter 4 above.



Outline of a typical improvement whiteboard with space for the most important targets, new ideas, the prioritisation matrix and the operational whiteboard. The improvement whiteboard is the focal point for ongoing departmental improvement work.

The most important targets are listed on the left to ensure visibility and proximity to the overarching goals of the organisation. Beneath them is space to add new ideas. It helps to maintain focus on the fact that improvement proposals mentioned are connected to the company's strategic direction.

The prioritisation matrix in the middle of the improvement whiteboard is used to assess the effect of the suggested idea together with

the expected effort required to implement it. The operational whiteboard is placed on the right.

An employee is given the responsibility for the ideas that have been launched and a status report is given at each meeting. This draws attention to the importance of following through on ideas and forces people to work in a structured way towards a result. The PDCA wheel can, as mentioned in chapter 4 above, be used to follow up on the ideas and ensure that they are correctly implemented.

To keep an eye on the department's ability to generate and execute new ideas, you should keep an ongoing register of the number of ideas carried out.

How to create your Lean whiteboards

To succeed with Lean whiteboards, you should do the following:

- Summon the entire department to a meeting where you can develop the whiteboards together based on the demands of management regarding overview, task distribution and direction.
- Assess whether it is necessary to have an operational whiteboard, an improvement whiteboard and a goal whiteboard, or whether all three can be combined in a sensible way.
- Use the department's list of daily and weekly tasks as your starting point.
- Agree a time, frequency and agenda for the meeting.
- Test if these work and adjust as necessary.
- Document the process so that everyone knows what is expected:
  - Who facilitates?
  - Who will substitute in the event of illness or holidays?
  - When the whiteboard was last updated, etc.?

Performance management whiteboard

A performance management whiteboard focuses on the KPIs that will show how well things are going with the process and the department. It gives an overview of the progress of the department's key metrics. It is used to ensure that employees are involved in the preparation and development of the key figures by which management direct operations. It is essential that they are simple to calculate and make sense to employees. The most preferable situation is where each employee's way of working has a measurable influence on output and thereby also on the key figures.

Outline of a performance management whiteboard, in this case showing that the organisation needs to focus on improving quality and costs.

A good performance management whiteboard is balanced between short- and long-term goals but should also show a clear prioritisation in relation to four overarching parameters: quality, lead time, costs and motivation. Sometimes it is necessary to prioritise which goals

must be realised first. Perhaps costs need to be higher than expected in order to ensure quality.

Manageability is created by adding a graph of the negative development of the key figures — red key figures indicate unsatisfactory performance — to the right of the relevant parameter. Conversely, satisfactory performance key figures are placed in the green column. Thus key figures are either satisfactory or unsatisfactory (something needs to be reassessed).

We recommend setting one key figure per graph and drawing the graph rather than using a program such as Microsoft Excel. Drawing the graph gives a more tangible and intimate picture of the reality.

Case 11: Six Sigma control charts used as performance management whiteboards

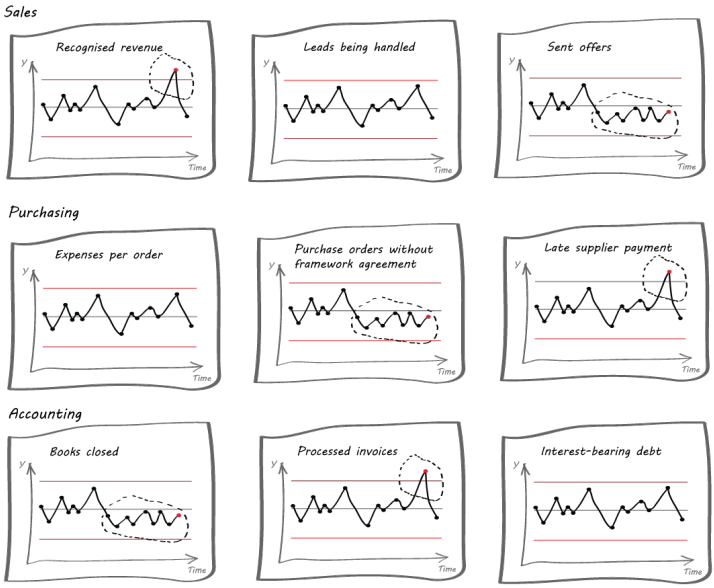
A Danish shipping company had worked for many years on optimising sales, purchasing and accounting processes. During that time it had worked with many different key figures to measure whether processes were developing correctly. Despite its best efforts, however, employees and managers found it hard to get a clear overview of the progress because many formats and methods for calculating key performance indicators (KPIs) were used across departments and countries.

We were asked to bring greater visibility and consistency across sales, purchasing and accounting processes using uniform formats. We chose to identify nine KPIs and illustrate them by using simple control charts.

As departments were spread across several countries, we constructed a website on which employees could display data by country or view a total for all countries.

For each process, we prepared an overview of the three main

process key figures and registered how the process continually evolved, as shown below. Each month we calculated new control limits.



Overview of the three main process key figures for sales, purchasing and accounts prepared for a Danish shipping company. This performance management whiteboard maintains organisational focus on the main check points.

Continuous improvement occurs by reacting to how the process evolves, finding the causes of the evolution and introducing improvements. Building a Six Sigma dashboard as a performance management whiteboard is a good example of how the elements of Lean and Six Sigma harmonise. Whiteboard meetings with a Lean perspective and control charts from Six Sigma combine to create an overview of the most central processes.

Whiteboards, for better or worse

Whiteboards can exercise enormous influence over an organisation. If they have been implemented correctly and are continually updated as appropriate, they can, in many cases, become the mainstay of a whole range of central management processes. They help to create an overview of important information such as the project status, the extent of knowledge-sharing between colleagues, project focus and task prioritisation, through making information readily available and easy to overview for everybody in the organisation.

The introduction of whiteboards and whiteboard meetings can make some employees feel that they are under surveillance and that the whiteboards are a sign of creeping control. That is why it is important to involve employees in the design of the whiteboards as well as ensuring that they understand that the purpose is not control but simply direction of everyday operations.

Many organisations employ people working in global teams at various locations making it a challenge to instil the right attitude about such whiteboards. However, there is a lot of software that supports digital Lean whiteboards without sacrificing the simple and intuitive character of the whiteboard (e.g. see <vimpl.com>).

Improvement as a participatory process

When you introduce continuous improvement you open the way for social decision-making. It becomes a democratic process to decide which improvement initiatives should be prioritised and in what order.

One pitfall is that suggested initiatives will be popular but will not have a great impact on results — things like a new coffee machine, cake on Fridays and a free massage. The proposals can also focus on problems in other departments and meetings can descend into criticisms of other parts of the organisation.

All of the issues mentioned may well be relevant. There may be

How to hold a great whiteboard meeting

A great whiteboard meeting takes between 15 and 30 minutes and contains status reports on:

- Daily operations: who is doing what today? What is the status since the last meeting?
- Target management: what do we measure? What is the latest measurement? Is a pattern evident in the key numbers?
- Improvement initiatives: which improvement ideas are we working on right now? Are there new ideas we should add to the whiteboard?

At fixed weekly meetings

Before the meeting:

- Consider who will facilitate the meeting, perhaps more than one person.
- Remind participants of the meeting.
- Make sure that everyone has updated the whiteboards.
- If possible, use five minutes before the meeting to get an overview of the status of the tasks so you know what the day's main topics will be.
- Be on time.

During the meeting:

- Do not be afraid of repeating what you said at the last meeting when summing up the current status.
- Allow each employee to speak.
- Make sure they do not spend too much time on the nitty gritty.
- Stick to the agenda.

- Conclude the meeting by listing the decisions you have come to.

After the meeting:

- Make sure the whiteboard is updated. Take a photo of the whiteboard as documentation and send it to managers or other stakeholders.
- Follow up on unanswered questions.
- Make sure that those employees who need to talk to each other about concrete tasks or issues schedule time for this.
- Make sure that your discoveries with regard to unanswered questions are communicated to relevant employees.
- Extract the main conclusions and reports and, if necessary, share them with management.

You can benefit from keeping an eye on how many employees come to the meetings, and what the mood is like during meetings. These are strong indicators of whether support for the meetings is declining.

problems in other departments, ergonomic desks might be desirable and cake on Fridays could well raise morale. If you choose to buy a new coffee machine, you can justify this by remembering that with simple quick wins, you demonstrate that Kaizen culture works, and you can focus on larger and more complex challenges later. Consider then setting rules whereby for the first three months, for example, all ideas must be implementable within the department, that no idea should cost more than €20, and that all ideas must have a measurable impact, even if it is small.

Initially, you will be bombarded with new ideas so set an upper

limit for the number of ideas that you want to handle at any one time. If the most you want is 10 ideas under implementation, then new ideas will have to wait until some of the older ones are concluded and implemented. This helps to increase focus on finalising improvements and achieving results. It works especially well in organisations where a lot of initiatives are launched but few are brought to conclusion.

At other times, you will find that no new ideas emerge at these meetings. Here are some suggestions to get the improvement process going again:

- Use a little more time at the ideas whiteboard at each meeting.
- Set targets whereby each employee must come up with three ideas before the next meeting.
- Go around to each desk distributing notebooks to employees. Ask each employee to note every time they see that some element of a process could be improved.

An essential part of the implementation of ongoing improvement is the work implementation itself, and it is extremely important that it remains simple and relevant for employees.

It is important that the improvements underpin the organisation's overall goals and the department's distinct objective. Therefore, it is a good idea to hang these objectives alongside the improvement whiteboard to allow you to refer to them when discussing improvement options.

The purpose of implementing a process of ongoing improvement is to ensure that you succeed in holding onto ideas, implementing them and ultimately producing results. Everyone in the organisation has a responsibility to always do things better. Everyone who identifies a problem or sees potential for improvement takes responsibility for it. The sum of all these small improvements can be enormous.

Over time, the organisation works itself closer to the ideal of *perfect processes.*

Summary

In this chapter, we have looked at process development and optimisation as part of daily operation. We introduced the chapter by highlighting the importance of creating a correlation between the development of the processes and the work in the processes. Furthermore, we gave an overview of an organisation where everyone was focused on improving the organisation's processes as a part of their work at all levels.

We sketched how whiteboards could help in visualising and directing work on the daily optimisation of processes. Operational whiteboards can be used to support ongoing processes. Improvement whiteboards can be used to collect, prioritise and implement improvement proposals. In addition, the target management whiteboard highlights the organisation's targets and performance — visible proof of the impact of the improvements.

Finally, we described ongoing improvement as a process, where a correlation between the organisation's targets and employees' everyday work needs are created. In this way, it becomes relevant for employees to contribute to the organisation's targets and to improve the organisation's processes on a daily basis. This allows the organisation to work towards *perfect processes* in the long term.

GLOSSARY

Tool	What is it?
Activities	Activities are the individual tasks that make up the completion of a process step.
Batch size	The amount sent through a process step at the one time in a 'batch', also called a 'series'.
Brainstorming	A group discussion where ideas are generated. The purpose of this is to facilitate the meeting, so that participants are inspired to think creatively and make suggestions as well as making sure that everyone has a say.
Business case	The calculation of the financial potential of a change as well as the costs of implementation. This is used as a benchmark for financial considerations and to compare and prioritise project alternatives in relation to the strategy.
Cause–effect matrix	Tool for creating an overview of Y factors and the prioritisation of X factors. Presented as a chart on which marks are given for each X factor in terms of its potential impact on a Y factor.
Changeover time	The time it takes to switch production to a new product.
Control charts	Control charts can be used to document the impact of the project, evaluate the success of the project, establish ongoing follow up and further optimise a process.
Control limit	Control limits define the area three standard deviations on either side of the mean of data.

Critical to quality (CtQ)	A translation of the customer's perception of quality to a measurable business parameter.
Data collection plan	Checklist of data definition, time and responsibility for data collection.
Operational definition	A description of what needs to be examined and the criteria under which data is collected. Everyone attributes the same meaning to concepts, minimising the risk of misunderstanding.
Data pilot	The collection of data for a short period, its evaluation and the collection of impressions and feedback from the data collectors.
Defects per million opportunities (DPMO)	Defects per million opportunities is a universal quality standard.
Fishbone (Ishikawa) diagram	A diagram that shows the causes of a specific event. It is a tool used to brainstorm X factors according to six aspects.
Gemba	The principle that to understand a process you must be present where the process takes place, 'where it happens'.
Lead time	The time it takes a particular component to get all the way through all the processes and all inventory stages. It is the sum of the processing time and the waiting time.
Lean house	An overview of the order of necessary improvements in a Lean implementation. Typically shown as a house or a temple illustration; it contains basic improvements (those needed to be carried out first) as its foundation with the more advanced improvements shown on top.
Milestone planning	Breaking the project down into milestones to provide a clear direction for the project and an opportunity to follow up on the success criteria along the way.
Normal distribution	A symmetrical distribution of observations around the mean.

Pareto analysis	Plots the frequency of a given task from frequent to seldom.
Plan–Do–Check–Act (PDCA) cycle	PDCA is a process for structuring ongoing improvement work.
PICK chart	A simple ranking of solution options set up on a two dimensional diagram with impact and investment/effort on the X and Y axes, respectively.
Process	A combination of steps performed in a particular order to produce something for a customer.
Process diagrams	An overview of a process, and the steps and role changes that form part of it.
Process steps	A collection of activities carried out by one department, which together transform several inputs into an output.
Project charter	A description of a project, what it must achieve and how, when and by whom; it is created by the project owners and major stakeholders. It is used to coordinate expectations, provide decision makers with a justification basis, coordinate responsibility, secure the necessary resources for the project and secure the strategic perspective.
Project organisation	An overview of who has what role in the project to ensure the involvement of the right people, harmonisation of expectations and responsibilities, and mobilisation and allocation.
Pull	Organising the cooperation between two process steps so that the second one determines when the first one is allowed to produce, thereby minimising inventory.

Push	Organising the cooperation between two process steps so that the first one determines when it produces, thereby potentially building up the inventory if the second step is not ready to use what has been delivered.
Quartiles	Classification of data in four parts with the same number of observations in each part.
Quick wins	Quick and easily implementable improvements which can be used to maintain the motivation for the project, show progress to stakeholders, achieve quick results and free up resources for the project.
Sigma rating	The scale from 0 to 6 sigma that indicates the deviation of process output compared to the output the customer expects. It is a universal standard for quality that permits the comparison of different processes and areas.
Single minute exchange of die (SMED)	Focuses on reducing the time used to change over between work on various cases or products.
Single piece flow	A tool, which requires that waste is eliminated to create flow whereby a case, order or item is forwarded to the next process as soon as it is ready for further processing.
Spaghetti diagram	Plot of a case's physical movement through a process; it is used if there is substantial transport time and the employee is in motion more than necessary.
Stakeholder analysis	An overview of who is affected by or who affects the project as well as their attitudes to and roles in the project.
Standard deviation (σ)	Expresses the observations' average distance to the mean.

Supplier, Input, Process, Output and Customer (SIPOC) method	A tool for a quick overview of the process inputs and outputs. Used to prioritise efforts, identify critical areas early on in the project and create a shared overview.
Swimlane diagram	A plot of responsibility handovers in the process.
Takt time	The time that elapses between each unit or series being completed within a time interval, for example, one day.
Task overview	An overview of the number and duration of the tasks, how they are prioritised, who can perform them, etc.
Value stream mapping	Plot of a single case's movement through and duration in various tasks from the initiating customer requirement to its fulfilment.
Voice of business (VoB)	An overview of the business' requirements of the process. Which things are critical to implement?
Voice of customer (VoC)	What we hear from the customer used to ensure that changes are based on the customer's requirements and priorities. The data is found via questionnaires, interviews, complaints, interactions, focus groups, etc.
Waste	Activities using resources on activities that are not creating value for customers or the company.

FURTHER READING

If you would like to read more on the topics covered in this book, we have compiled a list of recommended literature below.

Classical process optimisation literature

Books:

James P. Womack, Daniel T. Jones and Daniel Roos, *The machine that changed the world* (New York, 1990).
This book is widely seen as the first to give an account of Toyota's work on process optimisation in its production and to disseminate the ideas to a wider public. The book first introduced the five Lean principles.

Jeffrey K. Liker, *The Toyota way* (New York, 2004).
In the same way that Womack, Jones and Roos described the production processes, Liker identified the management principles underlying Toyota's systems not alone on the production floor but throughout the organisation.

Mike Rother and John Shook, *Learning to see* (Cambridge, MA, 2003).
If you would like to read more about value stream mapping, this is the book for you. It takes types of waste as its starting point and contains many examples and analyses.

John Seddon, *Freedom from command and control* (New York, 2005).
Lean tools transposed to service organisations with a strong focus on practical application and involvement.

Robert O. Martichenko, *Everything I know about Lean I learned in first grade* (Cambridge, MA, 2008).
An easy-to-read book about Lean that uses games and examples to show what Lean is — a learning and problem-solving culture.

Thomas B. Christiansen, Niels Ahrengot and Michael Leck, *Lean implementering i danske virksomheder* (Copenhagen, 2006).
A thorough examination of Lean with many concrete examples from Danish business.

The five most important articles about Lean from the *Harvard Business Review*:
James P. Womack and Daniel T. Jones, 'Beyond Toyota: how to root out waste and pursue perfection', *Harvard Business Review* (1996).
Steven Spear and H. Kent Bowen, 'Decoding the DNA of the Toyota production system', *Harvard Business Review* (1999).
Cynthia Karen Swank, 'Lean service machine HBR', *Harvard Business Review* [Add vol info.] (2003), [Add page range.].
Steven Spear, *Learning to Lead at Toyota*, *Harvard Business Review* [Add vol info.] (2004), [Add page range.].
Lean innovation

Allen C. Ward, *Lean product and process development* (Cambridge, MA, 2009).
Lean applied to product development.

Claus Sehested og Henrik Sonnenberg, *Lean innovation*, 2011).
Applying Lean to innovation processes with practical examples.

Eric Ries, *The Lean startup* (London, 2011).
Lean principles applied in innovative start-ups or projects.
Change management

John P. Kotter, *Leading change* (Boston, MA, 1996).
This book covers the foundation of organisational change with a practical eight step change model which is still used today as the backbone of numerous transitions.

Chip Heath and Dan Heath, *SWITCH* (New York, 2010).
The Heath brothers recognise in this book that change is difficult and review several of the psychological mechanisms in human nature that make it difficult for us to implement change.

Michael Beer and Nithin Nohria, *Breaking the code of change* (Boston, MA, 2000).
This book collects a series of articles from the most significant and renowned change theorists.